European Volatility Under the Microscope

A Machine Learning Approach to EURO STOXX 50 Forecasting

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Abstract

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Keywords: Volatility Forecasting, Realized Variance, High-frequency, EURO STOXX 50, HAR, Regularized Models, Tree-based Models, Neural Networks

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JEL Classification: C10, C50

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1 Introduction

The question of European financial autonomy, once confined to policy seminars, has in recent years become a central preoccupation of investors, regulators, and policymakers alike. In his landmark 2024 report on *The Future of European Competitiveness* European Commission (2025), Mario Draghi diagnosed the fragmented state of European capital markets as nothing short of an existential challenge, warning that without deep structural reform, including the formation of a Capital Markets Union aimed at creating a single integrated capital market across the European Union, Europe would fail to sustain its productivity, security, and long-term global competitiveness.

Against this backdrop, European equity markets have entered a new phase of elevated relevance to global investors. A structural refuge of capital outflows from US equities toward European capital markets accelerated in 2025, as rising trade policy uncertainty and historic depreciation of the US dollar prompted investors to diversify their geographical holdings. According to Morningstar (2026), European large-cap funds netted €41.2 billion of capital inflows in 2025, a stark increase from the €12.0 billion of outflows in 2024, while US large-cap funds closed with merely €6.2 billion on aggregate. In the longer run, UBS Flood (2025) projects that a base case reallocation of capital amounting to €1.2 trillion into European equities could materialize in the upcoming five years as investors divert there investments away from US equity markets. For all types of investors, both domestic and foreign, the ability to model, measure, and forecast the trajectory of European market risk has become an integral element of capital allocation, risk management, and long-term portfolio performance.

Volatility forecasting is the art of capturing and measuring the level of uncertainty prevalent across financial markets and instruments. More accurate and precise volatility forecasts translate directly into better calibrated risk models, resulting in more informed capital investments. The origins of the volatility forecasting literature trace back to the original works of Engle (1982) and Bollerslev (1986), who introduced the initial ARCH and GARCH models, respectively, to forecast volatility movements in financial markets. With time, access to high-frequency trading data and stronger compute allowed practitioners to expand this strand of literature, by introducing new specifications like the Heterogeneous Autoregressive (HAR) set of models originally derived by Corsi (2009). While the standard HAR models and its extensions have become salient benchmarks in volatility forecasting, advanced computers and programs have introduced advanced machine learning algorithms that can capture nonlinearities and complex patterns between correlated variables not previously observed by the conventional set of linear models in the forecasting literature.

Ongoing talks on forming a European Capital Markets Union, together with rising foreign capital infows to European markets, warrants a close examination of the volatility dynamics that prevail across European equity markets, along with understand which models are most capable of forecasting future volatility movements. The main objective of our paper is therefore to extend the analysis of Christensen et al. (2023) to the European context, by evaluating the out-of-sample volatility forecasting performance for a set of different models on the EURO STOXX 50 index. Importantly, our extension is motivated by three key considerations.

First of all, the empirical literature on ML-based volatility forecasting remains heavily concentrated on US equity markets, see e.g. Gunnarsson et al. (2024) and Mansilla-Lopez et al. (2025). To the best of our knowledge, no study has undertaken a comprehensive comparison between HAR-type models and modern ML approaches on the EURO STOXX 50 index. This gap in the literature is striking given the central role of the index as the primary benchmark for European equity markets performance, broadly comparable to the S&P500 in the US. Moreover, the EURO STOXX 50 index serves as the underlying asset for the VSTOXX index, Europe's principal measure of implied volatility, analogous to the VIX index for the S&P500 index in the US context.

Second of all, the macro-financial structure of the Eurozone is materially distinct from that of the US, with important implications for volatility dynamics and their predictability. Unlike the US, which constitutes a relatively integrated single-economy system, the EURO STOXX 50 index mirrors a multi-sovereign environment characterized by heterogenous fiscal conditions, sovereign risk differentials, and varying degrees of financial fragmentation, which have been shown to influence macroeconomic stability and asset price behavior (Lane (2012), Corsetti et al. (2013)). As a result, realized volatility in European equity markets is shaped not only by firm-level and aggregate factors, but also by supranational monetary policy conducted by the European Central Bank (ECB) (Bohl et al. (2008)), systematic financial stress as captured by the Composite Indicator of Systemic Stress (CISS) index (Hollo et al. (2012)), and heightened exposure to external shocks, including energy price fluctuations (Katsampoxakis et al. (2022)). Moreover, the open nature of the Eurozone economy implies that exchange rate movements and global spillovers from Asian and US trading sessions can play a more pronounced role in European equity markets relative to the US context (Wang et al. (2020), Golosnoy et al. (2015)). Taken together, these structural differences suggest that volatility forecasting models developed for US markets may not directly generalize to the European setting without explicitly accounting for these additional macro-financial variables highly relevant to the European context.

Finally, the single-index setting of our paper differs structurally from the cross-sectional approach of Christensen et al. (2023), whereby we study one target index series as opposed to the twenty nine individual stocks studied in Christensen et al. (2023). Moreover, contrary to Christensen et al. (2023), our cleaned sample spans a longer period from January 1999 to August 2020, thereby including the Dot-com Bubble (2000 - 2002), the Global Financial Crisis (GFC, 2008 - 2009), the European Sovereign Debt Crisis (Euro Crisis, 2010 - 2012), a prolonged low-volatility expansion period (2016 - 2019), and the Covid-19 market turmoil of March 2020. These five macroeconomic regimes provide a demanding and informative testing ground for volatility forecasting and model performance evaluation.

The paper is outlined as follows. Section (2) provides a historical and chronological overview of how the literature on volatility forecasting has evolved over time. Section (3) puts forth the data gathering and cleaning process while also motivating the selection of variables that we have choosen to include in our feature sets. Section (4) offers detailed insight into the functionalities of each and every forecasting model, as well as an explanation for the methodological approach we use to assess out-of-sample performance. Section (5) presents the empirical results gathered from the different feature sets we use in our approach. Finally, Sections (6) and (7) conclude our paper by discussing our findings, their implications, potential limitations of our paper while also exploring new avenues for future research on the topic of volatility forecasting.

2 Literature Review

This section reviews the literature on volatility forecasting by presenting a chronological overview of model development. We begin by introducing the autoregressive models, upon which modern research is derived from, and transition towards modern approaches based on high-frequency data, including linear HAR models and novel ML techniques. The section concludes by examining the existing empirical evidence for volatility forecasting on European equity markets and outlines how this study contributes to the existing strand of literature.

2.1 From ARCH Models to Realized Volatility

The origins of the literature on volatility forecasting can be traced back to the contribution of Engle (1982), who introduced Autoregressive Conditional Heteroskedastic (ARCH) models. This framework allows the conditional variance to vary over time as a function of past squared residuals, while the unconditional variance remains constant. Building on this foundation, Bollerslev (1986) extended the ARCH model to the generalized ARCH (GARCH) framework to allow for a more flexible and parsimonious lag structure in which the conditional variance depends on both past squared residuals and past conditional variances. As an infinite order ARCH model, the GARCH framework has been shown to outperform high-order ARCH specifications, thereby establishing itself as an important benchmark in volatility modeling.

Although widely applied in the risk management literature, standard GARCH models and their numerous extensions (see Lawrence R. Glosten (1993); Nelson (1991); Schwert (1990); Robert F. Engle (1993); Zakoian (1994); Sentana (1995); Hentschel (1995); Baillie (1996)), are typically estimated using daily return data. As a result, they disregard the vast amount of information contained in intraday price movements by treating each trading day as a single observation rather than as a series of smaller movements. This aggregation leads to a loss of information and limits the ability to fully capture the underlying volatility dynamics. Recognizing these limitations, Andersen (2001) proposed exploiting high-frequency intraday price data to construct realized volatility as a model-free and consistent estimator of the latent return variation. In contrast to parametric volatility models, realized volatility can be directly computed from observed price data, thus providing a more accurate measure of volatility ex-post. While typically highly skewed, Andersen (2001) show that the logarithmic transformation of realized volatility is approximately normally distributed and exhibits strong persistence, which makes it well-suited for standard time series modeling. Expanding on these insights, Andersen (2003) demonstrate that relatively simple linear models applied to realized volatility outperform traditional GARCH and stochastic volatility models in forecasting exchange rates. These findings highlight the importance of high-frequency data and mark a fundamental shift in the volatility forecasting literature, viewing volatility as something measurable rather than latent.

2.2 The HAR Model and Its Extensions

Following the seminal contributions of Andersen (2001) and Andersen (2003), which established realized volatility as a persistent and informative predictor of the latent volatility, Corsi (2009) introduced the heterogeneous autoregressive (HAR) model as a parsimonious and intuitive framework for volatility forecasting. The model is grounded in the Heterogeneous Market Hypothesis (HMH) of Ulrich A. Müller (1997), which asserts that financial markets consist of heterogeneous agents operating at different time horizons rather than a homogeneous group of investors, and it captures the long-memory behavior of realized volatility by approximating it through an additive cascade of volatility components measured over daily, weekly, and monthly horizons. Despite its simplicity, the HAR model effectively mimics the persistence observed in financial market volatility and it has been shown to outperform more complex autoregressive models with long lag structures.

Building on this framework, a large body of literature has proposed extensions to the standard HAR model to account for additional stylized facts of financial markets. One common extension is the logarithmic transformation (LogHAR), proposed by Corsi et al. (2012), which improves the statistical properties of realized volatility by reducing skewness and bringing it closer to a normal distribution. Other extensions focus on capturing more salient volatility dynamics. For instance, Corsi and Renò (2012) incorporate a persistent leverage effect into the HAR model (LevHAR), capturing the asymmetric effect of negative returns on future volatility. Similarly, Patton and Sheppard (2015) address potential asymmetries by proposing the semivariance HAR (SHAR) model, which divides realized volatility into its positive and negative return components. Their findings indicate that negative semivariance carries greater predictive power, highlighting that negative returns have a stronger and more persistent impact on future volatility than its positive counterpart. More recently, Bollerslev et al. (2016) extend the HAR framework by accounting for the time-varying measurement error in the realized volatility through the realized quarticity. The resulting HARQ model, which adjusts the parameters depending on the level of estimation uncertainty in the realized volatility measure, exhibits greater persistence during tranquil periods and faster mean reversion during turbulent periods, with meaningful improvements in forecast accuracy compared to the baseline HAR model with constant parameters. Taken together, these extensions demonstrate the flexibility of the HAR framework in capturing key features observed in financial data while maintaining a structure that is relatively easy to implement.

2.3 Macroeconomic and Financial Predictors of Volatility

A related strand of the literature examines the extent to which stock market volatility is driven by broader economic conditions, and whether the predictive performance of HAR-type models can by improved by incorporating additional sources of information beyond the lagged volatility components. This line of research complements the autoregressive nature of standard HAR specifications by emphasizing the role of macroeconomic and financial factors in governing volatility dynamics. In particular, Paye (2012) show that the persistence observed in realized volatility can be more accurately modeled by conditioning on macroeconomic and financial variables, suggesting that financial market volatility is not only driven by its own past but also reflects underlying economic fundamentals. This is in line with the findings in Charlotte Christiansen (2012), Robert F. Engle and Sohn (2013), and Nonejad (2017), where macroeconomic indicators play a significant role in predicting future volatility. Extending this argument beyond traditional macroeconomic variables, Yudong Wang (2018) demonstrate that crude oil volatility serves as a strong short-term predictor of stock market volatility. Their results suggest that augmenting autoregressive models with commodity-based measures can lead to meaningful improvements in forecast accuracy, further highlighting the importance of expanding the predictor set to consider a wide range of potential volatility drivers.

In addition to analyzing the impact of macroeconomic variables on financial markets, related studies explore the reverse relationship. For instance, Taufiq Choudhry (2016) investigate how stock market volatility and real economic activity across the US, Canada, Japan, and the UK are interrelated. Using a combination of linear and non-linear causality tests, they find that stock market volatility is a significant short-term predictor of the business cycle in all four countries, with stronger effects during periods of heightened market fluctuations. Moreover, their results document the presence of cross-country spillover effects, indicating that volatility and economic activity in the US can also influence economic fundamentals in other major markets. Collectively, this body of research suggests that the HAR family of models may omit relevant information by relying solely on lagged volatility components. As such, incorporating a broader set of macroeconomic and financial variables (hereafter referred to as the HAR-X model) can improve the predictive performance of these models, providing a strong motivation for conditioning the information set in forecasting applications.

2.4 Machine Learning Approaches to Volatility Forecasting

The application of machine learning (ML) methods in volatility forecasting is motivated by the limitations of traditional linear models when dealing with high-dimensional datasets, complex interactions among predictors, and potential non-linear relationships. As emphasized by Varian (2014), advances in data availability and computational power have made it possible to employ more flexible modeling techniques capable of capturing these patterns, which simple linear models tend to miss.

Traditional HAR models can be effectively extended by imposing regularization techniques, such as ridge regression, the least absolute shrinkage and selection operator (LASSO), and the elastic net. These methods address issues of overfitting and multicollinearity by selecting or shrinking the coefficients of variables, thereby allowing for more flexible model specifications with additional predictors compared to the rigid lag structure of the HAR framework (Hastie (2009a)). On this note, Audrino and Knaus (2016) investigate the ability of LASSO to recover the lag structure of the HAR model. They find that while LASSO can approximate the HAR model synthetically when it is assumed to be the true model, it struggles to do so when estimated on real data, suggesting that volatility dynamics may be more complex than implied by standard HAR models. In contrast, Yi Ding (2021) document that LASSO can outperform the HAR model in forecasting realized volatility, attributing this improvement in out-of-sample performance to its ability to model a more flexible lag structure. Similarly, Yu Wei (2020) show that the ridge regression can improve forecasting accuracy by mitigating multicollinearity and reducing overfitting in models augmented with additional predictors. In a related contribution, Francesco Audrino (2020) also show that sentiment and attention measures, such as social media, news article, and search engine data, can enhance predictive performance when combined with regularization techniques. Extending this line of research, Zhang et al. (2019) compare forecast combination approaches, which combine individual forecasts from HAR models, with shrinkage methods, including elastic net and LASSO, in predicting oil price volatility and find that shrinkage methods outperform both the combination approaches and the individual HAR forecasts.

Tree-based models constitute another class of models designed to capture non-linear relationships and interactions among predictors. In contrast to linear models, this framework adopts a non-parametric approach to partition the predictor set, thereby allowing for complex dependencies without requiring the functional form to be specified prior to estimation (Radmir Mishelevich Leushuis (2026). For instance, Yang et al. (2017) apply bagging techniques within the HAR framework to predict the volatility of commodity futures and find that augmenting HAR-type models with bagging methods leads to improved forecasting accuracy relative to the autoregressive benchmark. In related work, Chuong Luong (2018) employ random forests to high frequency data from the S&P ASX

200 and document improved predictive performance for the HAR model when this ensemble technique was applied. Similarly, Rangan Gupta (2022) show that random forests enhance the accuracy of forecasts for crude oil volatility, particularly at longer horizons. Building on this tree-based approach, Stefan Mittnik (2015) apply gradient boosting techniques and demonstrate that sequentially constructed decision trees, when used with a large set of financial and macroeconomic predictors, produce superior out-of-sample forecasts compared to GARCH-type benchmarks. Their findings provide further evidence that this class of models is able to capture the dynamics of realized volatility and the effects of macroeconomic and financial variables, which may influence volatility dynamics in a non-linear fashion.

Advanced research also explores the use of neural networks in volatility modeling, which offers a flexible approach to approximating complex and potentially non-linear relationships in volatility dynamics. Despite this non-parametric learning process, early contributions provide mixed evidence on their predictive performance relative to linear benchmarks. As a start, Marcelo Fernandes (2014) find no support that neural network-based models consistently outperform standard HAR specifications when forecasting the VIX, suggesting that increased model flexibility does not necessarily translate into improved predictive accuracy. On a similar note, Vortelinos (2017) compare neural networks with HAR, GARCH, and principal component-based approaches, and report only modest improvements from neural network compared to simpler, indicating that the long-memory property of realized volatility is most effectively captured by autoregressive linear benchmarks. In contrast, other studies document more favorable results. Miura et al. (2019) show that artificial neural networks can improve predictions of realized volatility in cryptocurrency markets, where volatility dynamics are highly non-linear subject to large breaks. Extending this argument, Bucci (2020) construct neural network models that incorporate a comprehensive set of macroeconomic and financial predictors and show that these models are able to outperform linear benchmarks, including the HAR model. Importantly, these results indicate that neural networks perform relatively better in forecasting during periods of financial distress, suggesting that realized volatility may display a stronger non-linear dynamics during times of heightened volatility and that this behavior is more accurately captured by flexible modeling approaches.

A comprehensive comparison of machine learning models and the linear HAR framework is provided by Christensen et al. (2023), which serves as the primary benchmark for this thesis. The authors evaluate the machine learning methods described above against the HAR family in forecasting realized volatility, examining both a concise set of lagged volatility components and an extended information set that incorporates firm-specific and macroeconomic variables. Their findings highlight that the relative performance of machine learning models is highly dependent on the information contained within the feature set. While traditional HAR-type models remain competitive in parsimonious settings, more flexible modeling techniques, particularly regularization methods and neural networks, deliver consistent improvements in forecasting accuracy when a large set of predictors is included. In contrast, while tree-based models exhibit weaker performance when only lagged volatility components are included, the predictive accuracy is increased when augmented with additional predictors, reflecting their ability to capture non-linearities and interactions.

These results are broadly consistent with the findings of Díaz et al. (2024), who conduct a similar evaluation of machine learning models and HAR-type specifications in forecasting S&P 500 realized volatility with a comprehensive set of macroeconomic and financial predictors. Their analysis confirms that machine learning approaches can enhance predictive performance, especially in a high-dimensional setting, while also emphasizing that the quality of the information set is crucial. In particular, they find that variables capturing financial and macroeconomic uncertainty have the greatest explanatory power in forecasting future volatility, suggesting that the effectiveness of machine learning models depends on both the structure of the underlying volatility dynamics and the availability of informative predictors.

2.5 Volatility Forecasting in European Markets

While a large body of literature has examined volatility forecasting in US equity markets, empirical evidence for European markets remains limited. Despite the increasing integration of European financial markets, relatively few studies focus on broad European equity indices and existing research in a European context tend to be fragmented in both scope and methodology. For example, Barry Harrison (2012) apply traditional AR and MA models to forecast volatility in Central and Eastern European markets, predominantly focusing on emerging economies rather than developed Western markets. In a different setting, Fabrizio Cipollini (2019) decompose realized volatility across five major European equity indices into common and idiosyncratic components and examine their contribution to EURO STOXX 50 volatility when integrated into a HAR framework. More recent work has explored the use of machine learning models in a European context. Javier Parra-Domínguez (2024) evaluate the ability of different neural network architectures to predict EURO STOXX 50 index levels. However, their focus on price prediction differs from the volatility forecasting application considered in this study, and does not address the performance of machine learning methods relative to established linear models.

Taken together, the existing evidence on European volatility forecasting is both sparse and fragmented, and to the best of our knowledge, no previous studies have systematically compared the HAR framework with novel machine learning approaches. Moreover, the role of macroeconomic and financial predictors in driving volatility dynamics in a European setting remains underexplored. In this light, this thesis provides a comprehensive comparison of linear and non-linear models in forecasting the realized volatility for the EURO STOXX 50 index, while incorporating a broad set of macroeconomic and financial variables into the modeling of stock market volatility.

3 Data

In this section, we provide an overview of the data gathering and preparation process that we conduct prior to forecasting realized volatility. We begin by exploring the data cleaning process, after which we proceed to present the training-validation-test split along with our two feature sets. We round-off this section by motivating the inclusion of each exogenous variable in our paper and we also present summary statistics of all variables included in the study.

3.1 Data Collection and Preparation

The empirical analysis in this paper is based on 1-minute intraday trading data for the EURO STOXX 50 index, obtained from the Stockholm School of Economics. Prior to data cleaning, the sample spans the period from January 1998 to August 2020. The EURO STOXX 50 index is composed of the 50 largest and most liquid companies in the Eurozone, and it represents a broad range of industries (see Appendix (8.1) for an industry and country composition). As such, it is widely regarded as the European equivalent to the S&P500 index and serves as a natural benchmark for assessing the performance of and volatility dynamics of European equity markets.

Prior to model estimation, we perform a comprehensive data cleaning process on the initial dataset, consistent with the existing literature. We begin by excluding weekends and other trading days with more than 50% missing intraday observations. In the initial sample, this results in 164 days being removed, which are distributed across the entire sample period. Importantly, these excluded trading days do not appear to be systematically concentrated in any particular period, suggesting that this data cleaning step is unlikely to introduce substantial survivorship bias to our dataset. Second, smaller data gaps are addressed by forward-filling missing prices within the trading day with the last available observation. Remaining missing values at the start of the trading day are backward-filled using the first available price within that day. We note that missing price data appears to be idiosyncratic and reflects data quality issues rather than systematic deficiencies. Overall, this approach ensures a complete intraday price series, preserving days with only minor interruptions while excluding those with insufficient data to support reliable realized variance estimation. We further describe the construction of the intraday trading window in Subsection (3.2).

After merging all variables into the extended feature set (see Subsection (3.4), the sample period is constrained by the availability of the VSTOXX index, which starts on January 5th, 1999. Consequently, the final dataset spans the period from 1999-01-05 to 2020-08-06 after the cleaning process. It is important to note that our analysis is based on the continuous EURO STOXX 50 index rather than fixed set of constituents. As a result, the data is subject to periodic index rebalancing, in contrast to Christensen et al. (2023) who focus on individual firms. While working with an index implies that the series captures the evolving structure of the European equity markets, it may also absorb discontinuities and jumps around rebalancing dates.

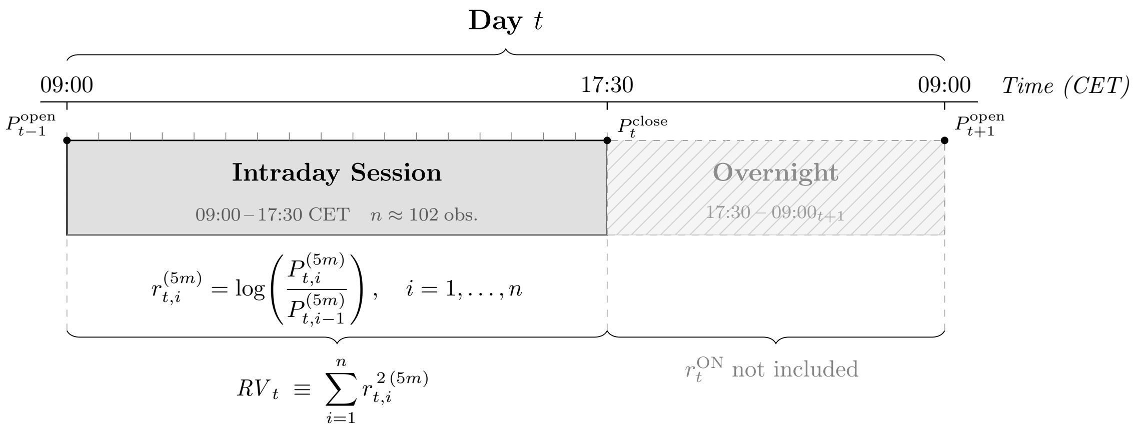
The final cleaned sample spans from January 5th 1999 to August 6th 2020 (5,489 trading days) and encompasses several distinct macroeconomic regimes, including the Dot-com Bubble (2000-2002), the Global Financial Crisis (GFC, 2008-2009), the European Sovereign Debt Crisis (Euro Crisis, 2010-2012), a prolonged low-volatility expansion period (2016-2019), and finally the Covid-19 market turmoil of March 2020. This wide range of market conditions provides a demanding and informative setting for evaluating volatility forecasting models, as it allows us to assess their performance during both tranquil periods and episodes of elevated market uncertainty.

3.2 Trading Window

The trading window of the EURO STOXX 50 is recorded between 09:00 to 17:30 CET, which is also confirmed by the official opening hours (STOXX Ltd. (2025)). When performing diagnostic checks on the intraday data, a sharp decline in trading activity is observed after 17:30, although some observations are still recorded after this mark. These post-close entries after likely to stem from closing auction mechanisms, index adjustments, recalculations, or data vendor artifacts rather regular trading activity. Given that the data is structurally different outside official opening hours, we restrict our analysis to the regular intraday trading window from 09:00 to 17:30. This approach is consistent with related studies, including Christensen et al. (2023).

While index levels are recorded at 1-minute intervals, we follow standard practice in the volatility forecasting literature and construct 5-minute log returns before proceeding with the realized variance computation. This adjustment mitigates the impact of microstructure noise associated with 1-minute intraday data, such as order imbalances, bid-ask bounces, and related market frictions, which can lead to biased volatility estimates. We illustrate the construction of the intraday trading window in further detail in Figure (1).

Figure 1: Intraday Trading Window and Realized Variance Construction



Notes: The figure illustrates the daily sampling window used to construct the realized variance (RV_t) estimator on day t for the EURO STOXX 50 index. The intraday session spans 09:00–17:30 CET, yielding $n = 102$ non-overlapping 5-minute log-returns per trading day. Using 5-minute intervals as opposed to 1-minute intervals is motivated to mitigate the adverse effects of market microstructure noise that inflate higher-frequency estimators. Note that the overnight return (r_t^{ON}) is excluded from the estimator, as it is driven by excluded from the estimator, as it is driven predominantly by non-trading-hours noise rather than a continuous-time price process. Section (4) for further information on this derivation.

3.3 Training-Validation-Test Split

Table (1) reports the sequential time-series split of the data into training (70%), validation (10%), and test sets (20%). The percentage split for each partition is in line with the broad forecasting literature, including Christensen et al. (2023). The training set is used for model estimation, the validation set for hyperparameter tuning where applicable, and the test set is reserved for out-of-sample performance evaluation.

Table 1: Training, Validation, and Test Set Partition

		Splits		
	Full Sample	**Training**	**Validation**	**Test**
Share of Sample	100%	70%	10%	20%
Observations	5,489	3,842	548	1,099
Start Date	1999-01-05	1999-01-05	2014-02-27	2016-04-21
End Date	2020-08-06	2014-02-26	2016-04-20	2020-08-06

Notes: The table reports the partition of the cleaned sample (5,489 trading days) into training, validation, and test sets. The training set is used for parameter estimation, the validation set for hyperparameter selection in the regularized linear models, gradient boosting, and neural network specifications, and the test set for out-of-sample forecast evaluation. For the non-regularized HAR benchmark models, bagging, and random forests, the training and validation sets are merged and parameters are re-estimated daily via a rolling window, consistent with Christensen et al. (2023).

3.4 Feature Sets

As discussed in previous sections, the choice of explanatory variables is a key determinant of predictive performance in the volatility forecasting literature. While the traditional HAR framework relies on a parsimonious set of lagged volatility measures realized over different time horizons, recent advances in machine learning have enabled the inclusion of a broader range of macroeconomic and financial variables to improve forecasting performance.

To examine how the information set influences forecasting performance, we follow Christensen et al. (2023) and construct two distinct feature sets. The first, denoted $\mathcal{M}_{\text{HAR}}$, contains only the core variables of the HAR framework, that is the daily, weekly, and monthly realized variance components. For model-specific extentions to the standard HAR model, such as realized quarticity (HARQ) and leverage effect components (Lev-HAR and SHAR), are included only in their respective model specifications. All models considered in this study, that is HAR benchmarks, regularized linear models, tree-based methods, and feed-forward neural networks, are first evaluated using only this restricted feature set. The second feature set, denoted $\mathcal{M}_{\text{ALL}}$, augments the baseline $\mathcal{M}_{\text{HAR}}$ set with a broader set of macroeconomic and financial predictors relevant for forecasting of realized volatility in a European setting. This expanded information set enables more flexible machine learning models to capture non-linear relationships and interactions among predictors, which the HAR lineage does not necessarily accommodate.

A comparison of forecasting performance across these two feature sets, ceteris paribus, allows us to investigate whether improvements in predictive accuracy are primarily attributable to increased model flexibility or the inclusion of additional informative variables.

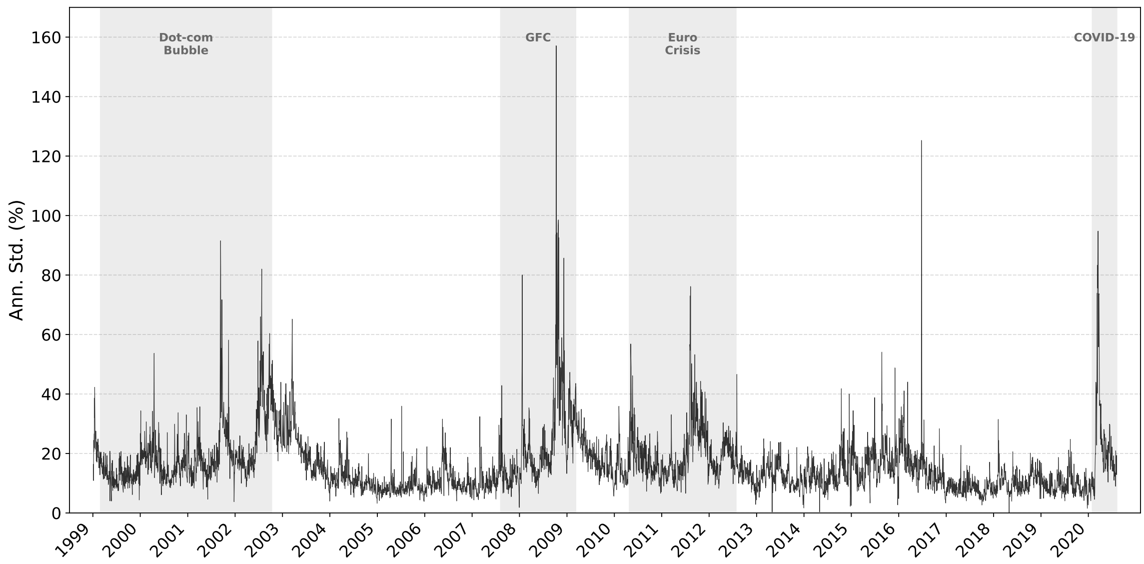
3.4.1 $\mathcal{M}_{\text{HAR}}$

As noted above, the $\mathcal{M}_{\text{HAR}}$ feature set consists of the standard HAR components, that is daily, weekly, and monthly realized volatility measures. In addition, it includes variables specific to each extension of the HAR framework considered in this study. For instance, the square root of realized quarticity enters the set as interaction term with realized volatility in the HARQ specification, but is not included in other models.

All models considered in this study are first evaluated using this restricted feature set. This setup enables a direct comparison between traditional HAR-type models and more flexible machine learning approaches when estimated on identical information sets, thereby isolating the effect of model complexity.

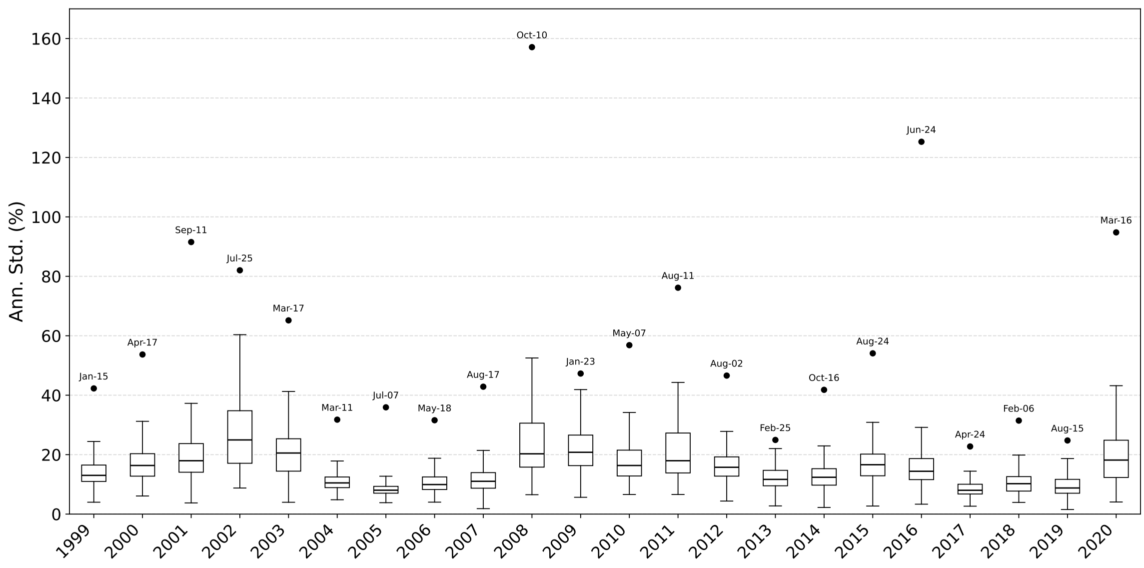
All variables used to construct $\mathcal{M}_{\text{HAR}}$ are derived from the same intraday trading dataset obtained from the Stockholm School of Economics. For illustrative purposes, the distribution of annualized daily realized volatility over the full sample period is presented in Figures (2) and (3).

Figure 2: Annualized Daily Realized Volatility (1999 - 2020)



Notes: The figure plots annualized daily realized volatility for the EURO STOXX 50 index over the sample period January 1999 - August 2020. Shaded regions indicate periods of elevated market stress identified as distinct volatility regimes: the Dot-com Bubble (March 1999 – October 2002), the Global Financial Crisis (GFC, August 2007 – March 2009), the European Sovereign Debt Crisis (Euro Crisis, April 2010 – July 2012), and the COVID-19 pandemic (February 2020 – August 2020). To annualize the numbers, we assume an average of 252 trading days per year.

Figure 3: Distribution of Annualized Daily Realized Volatility (1999 - 2020)



Notes: The figure illustrates a box plot of the annual distribution of daily realized volatility for the EURO STOXX 50 over the sample period January 1999 - August 2020. Each box displays the interquartile range of daily annualized realized volatility within a given year, with the horizontal line in the box denoting the median. The dot above each box marks the single highest volatility observation of the year, with the corresponding date indicated above. Note that the box for 2020 only includes data from January till August which marks the end of our sample period. To annualize the numbers, we assume an average of 252 trading days per year.

3.4.2 $\mathcal{M}_{\text{ALL}}$

While the $\mathcal{M}$HAR feature set is restricted to realized measures of volatility derived from high-frequency intraday data, the $\mathcal{M}$ALL feature set expands the information set by incorporating a broader range of financial and macroeconomic predictors. In contrast to $\mathcal{M}$HAR, which captures the endogenous persistence of volatility through its heterogeneous autoregressive structure, $\mathcal{M}$ALL incorporates exogenous sources of information that reflect forward-looking expectations, macroeconomic conditions, commodity price dynamics, as well as global market spillovers. As highlighted by previous studies focusing on exogenous drivers of stock market volatility, such an enriched information set allows the models to exploit a wider range of economic signals and potential non-linearities and interactions among predictors. This, in turn, provides a more comprehensive framework for forecasting future volatility in a European setting. The expanded set of predictors used in the $\mathcal{M}$ALL feature set is justified below.

1. Endogenous Volatility and Risk Expectations

The VSTOXX index measures the implied volatility of the EURO STOXX 50 index based on option prices with a 30-day maturity. As a forward-looking indicator of market-implied uncertainty, it captures the compensation investors demand for bearing volatility risk and is therefore widely used in the literature as a proxy for investor sentiment and market fear in European equity markets. The inclusion of the VSTOXX index is further motivated by the well-documented leverage effect and volatility feedback mechanism, whereby an increase in implied volatility is associated with higher realized volatility in subsequent periods (see Blair et al. (2001) and Bekaert and Wu (2000)). Moreover, Christensen et al. (2023) show that implied volatility to a large extent subsumes the information contained in historical realized volatility for forecasting purposes. In their analysis, the VIX index is included in levels as a predictor for future stock market volatility, which is in line with the specification adopted by Díaz et al. (2024). Building on this argument, Stefan Mittnik (2015) also identify implied volatility, measured by the VIX, as one of the primary drivers of realized volatility within the random forest framework. Accordingly, we include the VSTOXX index, sourced from the STOXX website, in levels format.

2. Macroeconomic and Systemic Risk Indicators

Following Christensen et al. (2023) and Díaz et al. (2024), we also include the Economic Policy Uncertainty (EPU) index as a predictor, originally developed by Baker et al. (2016). The index is constructed by counting the frequency of articles in major European newspapers that jointly contain terms related to economic policy and uncertainty in the Eurozone. These counts are normalized by the total number of articles to ensure comparability over time and across sources. Given that the index is available at a monthly frequency, we employ a one month lag to avoid look-ahead bias. The use of the EPU index as a predictor of realized volatility is further supported by the work of Francesco Audrino (2020), who show that indicators derived from social media, news article, and search engine data contain predictive information that can be used for improving forecasting accuracy. The data is obtained from the Economic Policy Uncertainty website.

In addition to the EPU index, we extend the set of macroeconomic and financial predictors by incorporating the Composite Indicator of Systemic Stress ($CISS$) index developed by Hollo et al. (2012) and provided by the European Central Bank (ECB). The CISS index is a daily measure of systemic risk in the European financial system, constructed by aggregating stress signals across multiple market segments, including equity, bond, money, and foreign exchange markets, as well as financial intermediaries. The index is derived from a range of market-based indicators, such as volatility measures, credit spreads, liquidity indicators, which are transformed into standardized stress measures and combined into a single composite index. Daily CISS data is extracted from the ECB website.

3. Exchange Rate Dynamics

We extend the framework of Christensen et al. (2023) by incorporating exchange rate dynamics through the EUR/USD exchange rate (EUR/USD), following the broader macro-financial predictor sets considered in Díaz et al. (2024) and Christiansen et al. (2012). In an open economy such as the Euro area, which is characterized by monetary and financial integration, exchange rate fluctuations capture changes in relative monetary conditions, global risk sentiment, and international capital flows. All of these factors can affect corporate earnings expectations, export competitiveness, and the valuation of internationally exposed firms, thereby influencing equity market volatility. Moreover, Fabrizio Cipollini (2019) show that European equity volatility is largely driven by underlying common components, which can be attributed to the interconnectedness across countries in the Euro zone. This suggests that variables capturing global economic conditions, such as foreign exchange rates, may contain relevant information for volatility forecasting. The EUR/USD exchange rate is sourced from the ECB and we compute its daily percentage change to account for short-term movements in the time series.

4. Commodity and Inflation-linked Factor

We further extend the set of exogenous predictors by including the European Brent crude oil price ($BRENT$) as a proxy for global commodity and inflation-linked shocks. This choice is motivated by the evidence in Yudong Wang (2018), who show that crude oil volatility contains significant predictive information for stock market volatility, particularly at short horizons. Their findings suggest that commodity price dynamics capture broader macroeconomic conditions, including inflationary pressure and global demand fluctuations, which are not fully reflected in traditional volatility models. In a European context, this measure is particularly relevant given the region's status as a net importer of energy and subsequent reliance on global commodity markets. As such, fluctuations in oil prices can influence both corporate earnings and perceptions of market uncertainty, which in turn have an effect on stock market volatility. We include the Brent oil price, obtained from the US Energy Information Administration (EIA), as an additional predictor in the extended feature set and calculate the daily change to capture commodity price shocks.

5. Global Market Spillovers

To account for global spillover effects, we extend the specification of Christensen et al.

(2023) by incorporating equity market returns from two major international markets. This is motivated by the central position of European markets within the global trading day, where information from earlier trading activity in the US and next-day trading in Asia may influence European volatility dynamics. The inclusion of spillover effects is also supported by Taufiq Choudhry (2016), who document strong cross-country effects between stock market volatility and economic fundamentals. Their findings highlight that financial market activity in the US can affect real economic activity in other parts of the world, suggesting that global market developments can play an important role in elevating stock market volatility in European markets.

We include daily returns from the Nikkei 225 index ($NIKKEI$) to account for spillovers from Asian markets, obtained from the Federal Reserve Economic Data (FRED) database. Note that these daily returns are predetermined relative to the opening of the European markets. All days corresponding to Japanese holidays, where there is now trading activity, we proceed to forward-fill this gap. Alongside the Nikkei 225 index, we retrieve daily returns for the S&P500 from the Wallstreet Journal Markets website ($S\&P500$). Note that contrary to the Asian markets, daily returns from the S&P500 are not predetermined relative to the opening of the European markets. Therefore, we lag the daily returns from the S&P500 by one day. Similarly to the approach for the Nikkei 225 index, we forward-fill all dates corresponding to American holidays where no trading takes place.

6. Monthly and Weekly Momentum

We also include and measure the weekly and monthly momentum of the EURO STOXX 50 index in our $\mathcal{M}_{\text{ALL}}$ feature set. While Christensen et al. (2023) only incorporates a weekly momentum variable to measure financial performance of each of the DJIA constituents in their study, we compute both weekly ($M1W$) and monthly ($M1M$) momentum factors or our single-index setting. Momentum is constructed from daily close prices as 5-day and 22-day cumulative log-price changes.

7. Term Spread and Credit Spread

To further capture expectations about future macroeconomic conditions, we extend the analysis of Christensen et al. (2023), who include the US 3-month treasury rate as a predictor, by incorporating measures of term structure dynamics and interbank credit risk. More specifically, we construct the term spread ($TERM$), measured as the difference between the 10-year euro interest rate swap and the 3-month EURIBOR, and the credit spread ($CREDIT$), approximated as the spread between the 3-month EURIBOR and the 3-month euro overnight index swap (OIS) rate.

The slope of the yield curve ($TERM$) is a well-established leading indicator of real economic activity (Estrella and Hardouvelis (1991), Estrella and Mishkin (1998)) and a recurring predictor in the volatility forecasting literature (Paye (2012), Christiansen et al. (2012), Conrad and Loch (2015)). The EURIBOR-OIS credit spread ($CREDIT$) is the euro-area analogue of the LIBOR-OIS (or TED) spread that Mittnik et al. (2015)) identify among the most informative predictors of US stock market volatility. Taken together, the two spreads provide complementary signals about monetary policy expectations and bank funding stress that a single short rate can not capture.

3.5 Summary Statistics

In the following section, we present summary statistics for all variables encompassed by the $\mathcal{M}_{\text{HAR}}$ and $\mathcal{M}_{\text{ALL}}$ feature sets respectively for the entire sample period spanning from the 5th of January 1999 to the 6th of August 2020. The summary statistics are presented in Table (2). Further information, particularly pertaining to the evolution of the macroeconomic predictors in the $\mathcal{M}_{\text{ALL}}$ feature set, are provided in Appendix (8.3).

Table 2: Summary Statistics for the $\mathcal{M}_{\text{HAR}}$ and $\mathcal{M}_{\text{ALL}}$ Feature Sets

	Mean	**Std. Dev.**	**Skew.**	**Exc. Kurt.**	**Min**	**Median**	**Max**
Panel A: $\mathcal{M}_{\text{HAR}}$ Feature Set							
RVD	16.14	9.96	3.17	20.33	0.15	13.76	157.11
RVW	16.51	9.33	2.62	11.04	4.52	14.18	93.58
RVM	16.83	8.75	2.15	6.46	5.29	14.40	71.78
RVD^{+}	11.22	6.79	2.77	13.51	0.09	9.54	83.33
RVD^{-}	11.37	7.66	3.80	32.94	0.12	9.50	138.64
RD^{-}	−0.44	0.80	−3.37	18.27	−10.54	0.00	0.00
RW^{-}	−0.21	0.38	−3.83	25.44	−5.41	0.00	0.00
RM^{-}	−0.10	0.19	−3.51	17.55	−1.91	0.00	0.00
RQ	0.65	19.50	50.39	2,681.48	0.00	0.01	1,129.75
Panel B: Additional Variables Added to Create $\mathcal{M}_{\text{ALL}}$ Feature Set							
$VSTOXX$	23.91	9.78	1.71	4.18	10.68	21.98	87.51
EUR/USD_{ret}	0.00	0.61	0.01	2.93	−4.74	0.00	4.20
$BRENT_{ret}$	0.03	2.68	−2.04	81.37	−64.37	0.03	41.20
EPU	4.93	0.45	0.00	−0.90	4.07	4.99	6.07
$CISS$	0.17	0.19	1.76	3.30	0.00	0.11	0.94
$NIKKEI_{ret}$	0.01	1.44	−0.38	6.81	−12.11	0.00	13.23
$S\&P500_{ret}$	0.01	1.25	−0.37	10.72	−12.77	0.06	10.96
$NIKKEI_{sq}$	2.09	6.19	12.73	248.63	0.00	0.49	175.15
$S\&P500_{sq}$	1.56	5.56	13.49	263.52	0.00	0.31	162.95
$M1W$	−0.00	3.12	−0.75	5.54	−27.07	0.24	16.47
$M1M$	0.02	6.20	−1.22	5.20	−47.60	0.76	22.34
$CREDIT$	0.20	0.26	2.88	10.33	−0.31	0.09	1.98
$TERM$	1.28	0.71	−0.04	−0.27	−1.00	1.25	2.90

Notes: This table reports summary statistics for the full sample covering the period from 5th of January 1999 to the 6th of August 2020 for both the $\mathcal{M}_{\text{HAR}}$ and the $\mathcal{M}_{\text{ALL}}$ feature sets respectively. The variables RVD, RVW, RVM, RVD^{+}, and RVD^{-} are shown as annualized volatility for interpretation purposes. Moreover, the variables RD^{-}, RW^{-}, RM^{-}, EUR/USD_{ret}, $NIKKEI_{ret}$, $S\&P500_{ret}$, $BRENT_{ret}$, $M1W$, $M1M$, $CREDIT$, and $TERM$ are shown as percentages. RQ is scaled by 10^6 while $NIKKEI_{sq}$ and $S\&P500_{sq}$ are scaled by 10^4. The $VSTOXX$ and $CISS$ are shown in levels, and the EPU index is shown as the one month lagged log value of the index. The $CREDIT$ is the 3-month EURIBOR minus the 3-month OIS rate, and the $TERM$ is constructed as the difference between the 10-year euro interest rate swap and the 3-month EURIBOR. See Tables (10) and (11) in Appendix (8.6) for further information.

4 Methodology and Empirical Design

In the following section, we outline the methodological framework and empirical design employed to generate the findings of our paper. We begin by introducing the concepts of realized variance and quarticity, after which we present the methodological structure underlying each forecasting model category and individual model.

4.1 Realized Variance and Quarticity

Following the literature on realized variance (see e.g. ?), the underlying price of an asset can be modeled as a continuous diffusion process, capturing the stochastic and time-varying nature of price movements over time. This continuous process can be according to Equation (1).

$$d\log(P_t) = \mu_t\, dt + \sigma_t\, dW_t \tag{1}$$

where P_t is the logarithm of the instantaneous asset price, μ_t is the long-term trend that varies with time, σ_t is the instantaneous volatility process, and W_t is a standard Brownian motion. The main objective of the volatility forecasting literature is to model the latent integrated variance. For a diffusion process, this is the integral of the instantaneous volatility over a specific interval (usually over a trading day) and it can be approximated following Equation (2).

$$IV_t = \int_{t-1}^{t} \sigma_s^2\, ds \tag{2}$$

While the true latent variance IV_t can not be directly measured, we estimate the integrated variance term using high-frequency intraday trading data. We partition each trading day into n equally spaced intervals of length Δ (so that $n = 1/\Delta$). Albeit we do have access to 1-minute high-frequency trading data, we follow the conventional strand of literature on high-frequency data motivation of and proceed to construct 5-min high frequency intervals to avoid microstructure noise, see e.g. Li and Xiu (2016). As a result, we define intraday log-returns according to Equation (3).

$$r_{t,i}^{(5m)} = \log\left(\frac{P_{t-1+i\Delta}^{(5m)}}{P_{t-1+(i-1)\Delta}^{(5m)}}\right), \quad i = 1, \ldots, n \tag{3}$$

As proposed by Andersen (2001), the latent integrated variance can be approximated by the sum of squared intraday returns when prices are sampled at a sufficiently high frequency. This estimator, referred to as realized variance, exploits the information in high-frequency data and provides a model-free estimate of the latent return variation. Realized variance can be defined according to Equation (4).

$$RV_t \equiv \sum_{i=1}^{n} r_{t,i}^{2\,(5m)} \tag{4}$$

Under standard regularity conditions, and as the sampling frequency shrinks ($\Delta \to 0$, equivalently $n \to \infty$), the realized variance is a consistent estimator of the latent integrated variance. However, in applications where the sampling occurs at finite frequencies, the realized variance measure is subject to estimation error. In particular, high-frequency limit theory provides a convenient representation of this by expressing realized variance as the sum of the latent integrated variance and an idiosyncratic error term (see Equation (5)):

$$RV_t = IV_t + \eta_t, \qquad \eta_t \mid IQ_t \sim \mathcal{N}(0,\ 2\Delta\, IQ_t)\,, \tag{5}$$

where the variance of the measurement error depends on the (latent) *integrated quarticity*, defined as in Equation 6)

$$IQ_t = \int_{t-1}^{t} \sigma_s^4\, ds. \tag{6}$$

The integrated quarticity measures the variation in volatility over time. In particular, it determines the magnitude of the estimation error in the realized variance estimate, where periods characterized by large fluctuations in intraday volatility (high IQ_t) lead to noisier estimates of IV_t whereas more tranquil periods (low IQ_t) imply more precise estimates. Given that IQ_t is not directly observable, it must be approximated. The realized variance literature traditionally use *realized quarticity* as a proxy for the latent variability measure, and it is outlined in Equation (7).

$$RQ_t \equiv \frac{M}{3} \sum_{i=1}^{M} r_{t,i}^4, \tag{7}$$

As before, the realized quarticity becomes a consistent estimator of the true integrated quarticity IQ_t as the sampling frequency increases. By relying on higher-order return moments, realized quarticity captures the intensity of intraday volatility fluctuations and provides a feasible, time-varying proxy for the uncertainty associated with realized variance. Consequently, RQ_t plays an important role in distinguishing between periods where realized variance is measured with high precision and periods where the estimator is subject to non-trivial noise.

4.2 HAR Benchmark Models

The linear benchmark models considered in our study are based on the heterogeneous autoregressive (HAR) framework. Building on the standard HAR specification, we also include a wide range of extensions that incorporate additional features observed in realized volatility dynamics, including the logarithmic transformation (LogHAR), leverage effect (LevHAR), semivariance decomposition (SHAR), and the time-varying measurement error (HARQ). These models form a uniform linear framework that augments different sources of volatility information while maintaining a parsimonious structure and therefore constitutes a natural benchmark against which more flexible machine learning approaches can be evaluated. We outline the specific model equations in more detail in the following subsections. The general linear equation for the standard HAR model and its extensions follows the structure of Equation (8), where Z_t denotes the feature set used to predict next day realized variance RV_{t+1}. The predictor vector $Z_t \in \mathbb{R}^p$ can thus be composed of the $\mathcal{M}_{\text{HAR}}$ or $\mathcal{M}_{\text{ALL}}$ feature sets respectively, as well as including the HAR-extension specific variables explained further in the sections below.

$$RV_{t+1} = \beta_0 + \beta' Z_t + \varepsilon_{t+1} \tag{8}$$

To evaluate the out-of-sample forecasting performance of the benchmark models, we adopt a rolling-window estimation approach following Christensen et al. (2023). Specifically, each model is estimated via ordinary least squares (OLS) using a fixed-length window corresponding to the combined training and validation sample, comprising of 4,390 observations, see Table (1). The estimation window is then rolled forward one day at a time across the test sample, and at each step, the model parameters are re-estimated using the information available up to time t to generate one-day ahead forecasts of realized variance, RV_{t+1}. We impose a non-negativity constraint on the forecasts to ensure economically meaningful predictions. This means that any negative forecasts are replaced by the minimum realized variance observed within the estimation window at time t, which is in line with the so-called 'insanity filter' implemented by Bollerslev et al. (2016). This procedure is commonly applied in volatility forecasting to prevent unrealistic negative variance predictions.

4.2.1 HAR Model

The Heterogeneous Autoregressive (HAR) model is a parsimonious time-series model designed to capture the long-term memory behavior of realized variance using a simple linear regression framework. It was introduced by Corsi (2009) to model high-frequency based volatility measures such as realized variance. As previously alluded to, the model is motivated by the Heterogenous Market Hypothesis (HMH), which argues that financial markets consist of agents operating at different investment horizons, such as short-term traders, medium-term portfolio managers, and long-term institutional investors.

Because these groups react to information over different time scales, volatility exhibits persistence across daily, weekly, and monthly horizons. The HAR model in Equation (9) captures this multi-scale structure by explicitly including volatility components measured at different aggregation levels, thereby providing a simple approximation to long-memory dynamics.

$$RV_{t+1} = \beta_0 + \beta_d RV_t + \beta_w \overline{RV}_t^{(w)} + \beta_m \overline{RV}_5^{(m)} + \varepsilon_{t+1} \tag{9}$$

In the equation above, RV_t denotes the realized variance observed at time t, while $\overline{RV}_t^{(w)}$ and $\overline{RV}_t^{(m)}$ represent the weekly and monthly averages of realized variance available at time t. These components are constructed as rolling means over the previous 5 and 22 trading days, respectively, thereby excluding non-trading days. All predictors are thus based on information available at time t and used to forecast the one-day ahead realized variance RV_{t+1}.

4.2.2 LogHAR Model

The LogHAR model, constructed by Corsi et al. (2012), is simply the logarithmic transformation of the original HAR specification, where the predictors are logarithms of the realized variance measures as opposed to levels (see Equation (10)). The LogHAR model is typically applied since realized variance is strictly positive, highly right-skewed, and exhibits strong heteroskedasticity. Estimating the model in logarithms stabilizes the variance of the series, reducing the influence of extreme volatility spikes.

$$\log(RV_{t+1}) = \beta_0 + \beta_d \log\left(\overline{RV}_t\right) + \beta_w \log\left(\overline{RV}_t^{(w)}\right) + \beta_m \log\left(\overline{RV}_t^{(m)}\right) + \varepsilon_{t+1} \tag{10}$$

When the HAR model is estimated in logarithms, forecasts are produced for $log(RV_{t+1})$, and given that the exponential function is nonlinear, we exponentiate the predicted value to generate downward-biased forecasts of RV_{t+1}. To account for this, we apply the standard log-normal bias adjustment seen in Equation (11).

$$\widehat{RV}_t = \exp\left(\widehat{\log(RV_t)} + \frac{1}{2}\widehat{\sigma}^2\right) \tag{11}$$

4.2.3 LevHAR Model

While the standard HAR model captures volatility dynamics across multiple horizons, it does not account for the well-documented asymmetry whereby negative returns have a larger impact on future volatility than positive returns of similar magnitude. This phenomenon, commonly referred to as the leverage effect, implies that adverse market movements are associated with stronger and more persistent increases in stock market volatilty. To account for this asymmetry, Corsi and Renò (2012) extend the standard HAR framework by including additional leverage terms, as shown in Equation (12). These terms augment the model with aggregated negative return components, allowing the LevHAR model to assign greater weight to downside risk

$$RV_{t+1} = \beta_0 + \beta_d RV_t + \beta_w \overline{RV}_t^{(w)} + \beta_m \overline{RV}_t^{(m)} + \underbrace{\gamma_d RD_t^- + \gamma_w RW_t^- + \gamma_m RM_t^-}_{\text{Leverage effect}} + \varepsilon_{t+1} \quad (12)$$

The additional leverage components are constructed from negative returns. We define $r_t^- = \min\{0, r_t\}$ as the negative part of daily returns, and set $RD_t^- = r_t^-$. The weekly and monthly leverage terms, RW_t^- and RM_t^-, are then computed as averages of the negative returns over the past 5 and 22 trading days, respectively.

4.2.4 SHAR Model

Expanding on the asymmetric impact of positive and negative returns on stock market volatility, the semivariance HAR (SHAR) model proposed by Patton and Sheppard (2015) decompose realized variance into its positive and negative semivariance components, allowing volatility to respond differently to upside and downside variation, as shown in Equation (13). Specifically, realized variance can be expressed as the sum of these components, $RV_t \equiv RV_t^+ + RV_t^-$, where RV_t^+ and RV_t^- denote the positive and negative semivariances, constructed as the sum of intraday squared returns conditional on returns being positive and negative, respectively.

$$RV_{t+1} = \beta_0 + \beta_d^+ RV_t^+ + \beta_d^- RV_t^- + \beta_w \overline{RV}_t^{(w)} + \beta_m \overline{RV}_t^{(m)} + \varepsilon_{t+1} \quad (13)$$

4.2.5 HARQ Model

The HARQ model, introduced by Bollerslev et al. (2016), extends the standard HAR framework by explicitly accounting for the time-varying measurement error inherent in realized variance. As a non-parametric estimate of latent volatility, realized variance is subject to estimation noise whose variance is proportional to integrated quarticity. Ignoring this feature may lead to an errors-in-variables problem, which in turn induces attenuation bias in the estimated coefficients. The HARQ extension addresses by augmenting the standard HAR model with an interaction term between daily realized variance and the square root of realized quarticity, as evident in Equation (14). This interaction allows the the coefficient on daily realized variance to vary over time so that the contribution of lagged realized variance is higher when this measurement error is low, whereas periods of high measurement error lowers the impact.

$$RV_{t+1} = \beta_0 + \beta_d RV_t + \beta_{dQ} \left(RV_t \sqrt{RQ_t} \right) + \beta_w \overline{RV}_t^{(w)} + \beta_m \overline{RV}_t^{(m)} + \varepsilon_{t+1} \quad (14)$$

In the equation above, RQ_t represents realized quarticity at time t, which serves as a proxy for the variability of the measurement error in RV_t. The interaction term $RV_t\sqrt{RQ_t}$ captures how said measurement error affects the predictive power of realized variance. Following Bollerslev et al. (2016), the quarticity adjustment is only applied to the daily lag of realized variance. This is because the measurement error is more pronounced at higher frequencies, whereas temporal aggregation over weekly and monthly horizons reduces much of the noise. Consequently, additional quarticity adjustments for these components will not be necessary.

4.2.6 HAR-X Model

We complete the set of linear benchmark models with the HAR-X specification, following Christensen et al. (2023). The HAR-X model extends the standard HAR framework by augmenting the realized volatility components with additional macroeconomic and financial predictors. Formally, the HAR-X model is given by Equation (15), where X_t denotes the vector of additional predictors used in the $\mathcal{M}_{\text{ALL}}$ feature set and γ is the corresponding coefficient vector. As such, the HAR-X specification collapses to the standard HAR model in the $\mathcal{M}_{\text{HAR}}$ feature set when no exogenous predictors are included.

$$RV_{t+1} = \beta_0 + \beta_d RV_t + \beta_w \overline{RV}_t^{(w)} + \beta_m \overline{RV}_t^{(m)} + \gamma' X_t + \varepsilon_{t+1} \tag{15}$$

4.3 Regularized Linear Models

Regularized linear models are a set of extensions of the standard linear regression (OLS) models that incorporates a penalty term to control model complexity and prevent overfitting. The general loss function for a regularized linear model can be written according to Equation (16), illustrated below.

$$\min_{\beta_0,\beta} \left\{ \sum_{t=1}^{T} (y_t - \beta_0 - X_t\beta)^2 + \lambda \cdot \phi(\beta) \right\} \tag{16}$$

In this setting, the first component of the expression, the sum of squared residuals $\sum_{t=1}^{T}(y_t - \beta_0 - X_t\beta)^2$, corresponds to the standard ordinary least squares loss function and measures how well the model fits the observed data. In the context of this paper, y_t denotes the the realized variance at time t, while X_t represents the vector of predictors, such as the lagged realized variance measures and the additional explanatory variables included in the extended feature set $\mathcal{M}_{\text{ALL}}$. Minimizing this first term of the equation would yield the OLS estimator, which is optimal in-sample but prone to overfitting when the number of predictors is large or when multicollinearity is present, both of which are prevalent in high-frequency volatility forecasting.

To address this issue, the model incorporates a penalty term $\lambda \cdot \phi(\beta)$, where $\phi(\beta)$ is a function of the regression coefficients and $\lambda \geq 0$ is a tuning parameter controlling the strength of regularization. Note that if $\lambda = 0$, the regularized linear model is the same as the standard linear regression model. This term thus imposes a cost on large coefficients, effectively shrinking them towards zero. The inclusion of this penalty introduces bias into the estimate but reduces their variance, with the objective to improve out-of-sample performance.

Different versions of $\phi(\beta)$ give rise to different regularization models. For instance, a quadratic penalty term corresponds to the ridge regression (RR), an absolute value penalty terms leads to the LASSO (LA) regression, and a combination of both RR and LA yields the elastic net (EN). In the following subsections, we follow the approach of Christensen et al. (2023) and present these three regularized linear regression models used in our study.

Before proceeding, to the model specific specifications, we summarize the methodological choices that are held constant across the three regularized models, so that any difference in out-of-sample performance is only attributable to the penalty structure and the selected feature set.

First, All predictors are standardized to zero mean and unit variance using moments computed from the rolling training portion of each window only, with standardization parameters re-estimated at every forecast origin. This prevents look-ahead bias and ensures that the penalty term operates on a comparable scale across coefficients, which is essential for the validation of L_1, L_2, and combined penalty terms.

Second, the out-of-sample forecasts are produced using a fixed-length rolling window of size $n_{\text{train}} + n_{\text{val}} = 4{,}390$ observations, see Table (1). At each forecast origin, the window slides forward by one trading day, the hyperparameters are re-tuned on the validation portion of that window, and a one-day-ahead forecast of realized variance is produced. Importantly, the model is refit on the rolling training portion only, using the selected hyperparameter(s) from the validation set. As such, no refit on the union of the training and validation set is performed prior to forecasting, consistent with the approach presented in Christensen et al. (2023).

Third, hyperparameter selection is captured by minimizing validation MSE over a logarithmically spaced grid of penalty parameters. For each candidate value (or, in the case of the EN, each combination of values), the model is estimated on the rolling training portion and evaluated on the rolling validation portion. The set of values minimizing the validation MSE is then used to refit the model on the rolling training portion and produce the one-day-ahead forecast of realized variance.

Lastly, since realized variance is a strictly non-negative metric, we apply an insanity filter similarly to that of Bollerslev et al. (2016) to all three models. Any forecasted value that falls below the minimum realized variance observed in the corresponding rolling training window is replaced by that minimum. This specification ensures that the regularized linear forecasts remain economically meaningful.

4.3.1 Ridge Regression

The Ridge Regression (RR), originally crafted by Hoerl and Kennard (1970), aims to reduce variance of the model by adding a quadratic penalty term to shrink all coefficients continuously toward zero, but rarely setting them exactly equal to zero. This is known as the L_2 penalty term and thus penalizes large coefficients more heavily. The penalty term is detailed in Equation (17).

$$\phi(\beta) = \sum_{j=1}^{p} \beta_j^2 \tag{17}$$

Adopting this technique can prove to be highly effective in improving volatility forecasting with feature sets that are composed of many variables that are highly correlated and where a simple linear regression model risks overfitting noise.

To determine the appropriate degree of regularization for the RR model, the ridge penalty parameter λ is selected using validation-based grid search. We consider a logarithmically spaced grid of candidate values defined as $\lambda_k \in \{10^{-5}, \dots, 10^{5}\}$ with $K = 1000$. This construction ensures that coverage of both near OLS and heavily regularized models.

4.3.2 LASSO Regression

The LASSO (LA) regression is another form of regularized linear model introduced by Tibshirani (1996). Unlike the ridge regression, the LA regression instead adopts what is commonly known as the L_1 penalty term to the original OLS regression, which sums the absolute value of the coefficients.

$$\phi(\beta) = \sum_{j=1}^{p} |\beta_j| \tag{18}$$

With this alternative set-up for the penalty term presented in Equation (18), the loss function of the LA regression can effectively shrink coefficients to zero, effectively performing variable selection.

The LA penalty parameter λ is, similarly to the RR model, determined via a validation-based grid search. In this case, we consider a logarithmically spaced grid of candidate values given by $\lambda_k \in \{10^{-10}, \dots, 10^{1}\}$ with $K = 1000$, allowing the model to approximate the unregularized OLS solution for sufficiently small values while also spanning levels of penalization that may induce sparsity.

4.3.3 Elastic Net

The Elastic Net (EN) regression, derived by Zou and Hastie (2005), can be interpreted as a combination of the RR and LA regression, designed to overcome the individual shortcomings of each of these models. While the RR retains all predictors but fails to perform variable selection, and the LA regression performs variable selection but struggle with stability when predictors are highly correlated, the EN combines both the L_1 and L_2 penalty terms respectively into a single penalty function. This penalty term is captured by the relationship in Equation (19).

$$\phi(\beta) = \alpha \sum_{j=1}^{p} |\beta_j| + (1-\alpha) \sum_{j=1}^{p} \beta_j^2 \tag{19}$$

In Equation (19), $\alpha \in [0, 1]$ is the mixing parameter governing the contribution of each penalty term. Assigning $\alpha = 1$ yields the LA model, while setting $\alpha = 0$ produces the RR model, such that the EN nests both models as special cases.

The EN specification introduces two hyperparameters. The first one is the overall penalty parameter λ, controlling for the strength of the regularization, and the second one is the mixing parameter α, determining the relative contribution of the L_1 and L_2 components respectively. Both parameters are selected using a validation-based grid search. Specifically, we consider a logarithmically spaced grid for the penalty parameter defined as $\lambda_k \in \{10^{-5}, \ldots, 10^2\}$ with $K = 1000$, alongside a disrete grid for the mixing parameter $\alpha \in \{0.01, 0.05, 0.10, 0.20, 0.30, 0.40, 0.50, 0.70, 0.90, 1.00\}$. This scope allows the EN to span specifications ranging from ridge-dominated shrinkage to LASSO-like variable selection.

4.4 Tree-based Models

Tree-based models belong to another set of frameworks well-suited for volatility forecasting because the characteristics of these models take into account the various features of financial data, namely nonlinearity, interaction effects, and high-dimensional data. Therefore, tree-based models address many of the core limitations of conventional linear models.

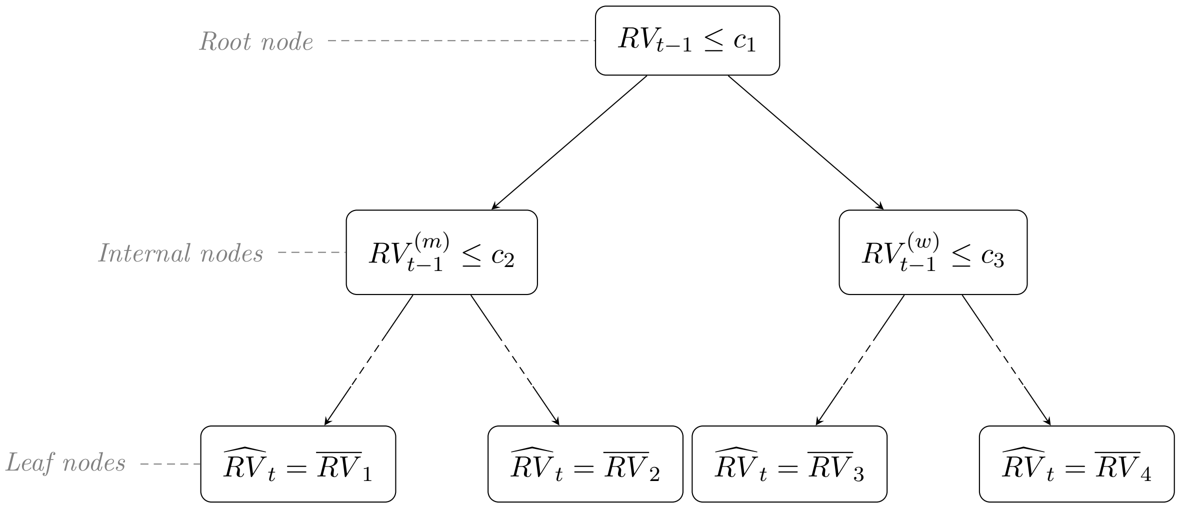
A single decision tree is a non-parametric machine learning approach that makes predictions from the data by recursively splitting the data into subsets based on feature values and criteria, forming a tree-like structure of decision rules. Predictions are made by averaging the values that end up in the terminal nodes of the tree, known as leaf nodes. Figure (4) serves as an illustratory example of a single decision tree.

Similarly to the approach presented in Christensen et al. (2023), we apply three well-documented volatility forecasting tree-based models in our paper. These are; Bagging (BG) Breiman (1996), Random Forest (RF) Breiman (2001), and Gradient Boosting (GB) Friedman (2001).

4.4.1 Bagging

Single decision trees have a static structure, and because of this characteristic single decision trees are prone to overfitting, inherently unstable, and subject to high variance. To address these shortcomings, Breiman (1996) proposed a bootstrapped aggregation technique, whereby multiple independent trees are built and estimated on resampled subsets of the original data and subsequently averaged to make a prediction. All subsets of the data are equally large and randomly formed, including replacement. Accordingly, the BG forecast is constructed as the average prediction across B regression trees, following

Figure 4: Illustration of a Regression Tree for Realized Variance



Notes: The figure illustrates a stylized regression tree using $\mathcal{M}_{\text{HAR}}$ predictors. Each internal node partitions the predictor space based on threshold values c_1, c_2, c_3, and each leaf node reports the average realized variance $\overline{RV}_j$ within that region where j is the total number of leaf nodes in the tree. The dashed segments along the second-level edges indicate that additional splits may occur before reaching the leaf nodes.

Equation (20).

$$\hat{f}(Z_t) = \frac{1}{B}\sum_{b=1}^{B} \hat{f}^{(b)}(Z_t) \tag{20}$$

In the equation above, $\hat{f}(Z_t)$ denotes the prediction from the BG model and $\hat{f}^{(b)}(Z_t)$ represents the prediction from the bth bootstrapped sample, i.e. the bth tree. Moreover, Z_t is the feature set and B is the total number of trees in the BG model.

In accordance with Christensen et al. (2023), the BG model is implemented in an off-the-shelf manner with minimal hyperparameter tuning. The base learner is a regression tree with a fixed minimum leaf size of 5, see e.g. Hastie (2009b), and the ensemble consists of $B = 500$ trees. As the BG model does not require any hyperparameter tuning, the validation sample is not used for model selection but instead combined with the training sample to form the estimation window.

Finally, to ensure economically meaningful predictions, any negative realized variance forecasts from the model are replaced with the minimum realized variance observed within the corresponding estimation window, following the same consideration as Christensen et al. (2023).

4.4.2 Random Forest

Random Forests (RF), introduced by Breiman (2001), is a machine learning method designed to improve predictive out-of-sample performance by combining and averaging a large number of regression trees. The approach builds directly on the concept of bootstrap aggregation (bagging, BG). While BG reduces variance by averaging across independently estimated trees, RF extends this framework by introducing an additional layer of randomness through feature sub-sampling, with the objective to further decorrelate individual trees and enhance out-of-sample performance.

Let $Z_t \in \mathbb{R}^p$ be the feature set, i.e. the predictor vector. The RF predictor is defined in accordance with Equation (21).

$$\hat{f}(Z_t) = \frac{1}{B}\sum_{b=1}^{B} \hat{f}^{(b)}(Z_t, \theta^{(b)}) \tag{21}$$

In this equation, B denotes the number of trees used in the forest and $\hat{f}^{(b)}(\cdot)$ is the prediction stemming from the b-th regression tree. The only characteristic that make the RF model distinct from BG is the $\theta^{(b)}$ notation, which embodies the random set of feature variables drawn at each split. Each tree partitions the predictor space into a set of disjoint regions, so called leaf nodes, and assigns a constant prediction within each region. Specifically, a regression tree can be written as illustrated in Equation (22).

$$\hat{f}^{(b)}(Z_t) = \sum_{m=1}^{M_b} c_m^{(b)}\, \mathbf{1}\big\{Z_t \in R_m^{(b)}\big\} \tag{22}$$

Here, $\left\{R_m^{(b)}\right\}_{m=1}^{M_b}$ represents the partition of the feature space created by tree b, and $c_m^{(b)}$ is the average of the target variable within region $R_m^{(b)}$. Similarly to the implementation of the BG model, we set the fixed minimum leaf size to 5 (see Hastie (2009b)) and the number of trees in the ensemble to $B = 500$.

The specification for the RF model relies on two critical sources of randomness. First, as previously alluded to, each tree is estimated on a bootstrap sample drawn with replacement from the original data set. Let $\mathcal{D} = \{(x_t, y_t)\}_{t=1}^{T}$. For each tree b, a bootstrap sample $\mathcal{D}^{(b)}$ is formed, introducing variability across trees and reducing the tendency of individual decision trees to overfit the data.

Second, and more importantly, RF differs from standard BG by introducing random feature sub-sampling at each split in the tree. Rather than considering all p predictors in the feature set Z_t when determining the optimal split, only a randomly selected subset of size m predictors is evaluated. In this paper, we follow the conventional recommendation for regression forests and define m according to Equation (23).

$$m = \left\lfloor \frac{p}{3} \right\rfloor \tag{23}$$

This specification is well established in the literature on RF (Breiman (2001), Hastie (2009b)) and serves to reduce the correlation between trees. Intuitively, if all predictors were considered at each split, dominant explanatory variables, such as daily lagged realized variance, would be repeatedly selected across trees, resulting in highly similar and correlated tree structures. By restricting the candidate set of predictors for each tree b

according to Equation (23), the RF model forces individual trees to explore alternative splits, thereby increasing model diversity and improving the effectiveness of averaging.

For volatility forecasting, this behavior is particularly useful, as volatility dynamics are known to exhibit nonlinearities and complex interactions across time horizons and macro-financial variables, see Christensen et al. (2023). RF can flexibly capture such patterns without requiring explicit specification of interaction terms or functional forms.

Since, similarly to the BG model, RF does not rely on hyperparameter tuning, the training and validation windows are combined to form basis for the estimation window. Moreover, any negative realized variance forecasts are subject to a non-negativity constraint, whereby these values are replaced with the minimum realized variance observed within the corresponding estimation window, similarly to the application of the BG model.

4.4.3 Gradient Boosting

Gradient Boosting (GB) is a tree-based machine learning algorithm originally proposed Friedman (2001). Rather than fitting a single complex model, GB constructs and ensemble of shallow decision trees sequentially, where each tree is trained to correct the errors of the preceding ensemble. This iterative refinement procedure is grounded in the principle of functional gradient descent, where the negative gradient of the loss function with respect to the current prediction serves as the target for each new tree.

Formally, the GB algorithm is initialized with a constant prediction as described by the equation below.

$$\hat{f}^{(0)}(Z_t) = \arg\min_{\beta_0} \sum_{t \in \text{train}} (RV_t - \beta_0)^2, \tag{24}$$

Under the mean squared error loss function, the model reduces the sample mean of realized variance. At each subsequent iteration $b = 1, \ldots, B$, a shallow regression tree fitted to the residuals of the current ensemble, and the prediction is updated according to Equation (25).

$$\hat{f}^{(b)}(Z_t) = \hat{f}^{(b-1)}(Z_t) + \nu \sum_{j=1}^{J_b} \hat{\gamma}_{jb} \mathbf{1}_{\{Z_t \in R_{jb}\}}, \tag{25}$$

Here, R_{jb} denotes the j-th terminal node of the b-th tree, $\hat{\gamma}_{jb}$ is the optimal step size with that node, and $\nu \in (0, 1]$ is a learning rate that controls the contribution of each tree to the final prediction. The final forecast is provided by the accumulated ensemble $\hat{f}(Z_t) \equiv \hat{f}^{(B)}(Z_t)$.

Each individual tree in the ensemble constitutes a so-called weak learner, a model only marginally better than a naive benchmark, but their sequential combination produces powerful nonparametric forecasting model capable of capturing nonlinearities and interaction effects in the predictor space without requiring these to be specified by the researcher *a priori*.

A well-known limitation of GB in financial applications is its sensitivity to outliers. Because the algorithm places disproportionate weight on large residuals when fitting each successive tree, extreme observation, such as volatility spikes during market stress episodes, can distort the learned function. This stands in contrast to Random Forests, which mitigates this issue through independent bootstrap sampling. Given this pronounced right-skewness and heavy tails of realized variance, this sensitivity is a relevant consideration in our setting, as noted by Christensen et al. (2023).

Following Christensen et al. (2023), the GB model is estimated using a rolling window scheme, where each forecast origin the hyperparameters are re-selected on the validation portion of the current window. Specifically, the tree depth is tuned over $\{1, 2\}$, the learning rate over $\{0.01, 0.1\}$, and the number of trees over $\{50, 100, \dots, 500\}$, with all remaining parameters set to their default values as proposed in the original implementation. The combination of minimizing the mean squared error on the validation set is retained, and the model is re-fitted on the full training window prior to generating the one-day-ahead forecast. Any negative forecast is replaced by zero, following standard practice in the realized variance literature.

4.5 Neural Networks

To complement the previously presented set of models, we build a family of feed-forward neural networks (NNs) to forecast next-day realized variance. Unlike conventional simple linear regression models, NNs are capable of approximating highly nonlinear and complex functional relationships between predictors and the response variable. We adopt the same four NN architectures as Christensen et al. (2023), denoted as NN_1, NN_2, NN_3, and NN_4 respectively. See Figure (5) for details on the respective architecture of each neural network. Moreover, across these four models, the only intended methodological difference is the depth and width of the hidden layer structure. This implies that all pre-processing, optimization, regularization, and forecasting technique are held fixed.

4.5.1 Feed-forward Neural Network Architectures

A feed-forward neural network maps the predictor vector available at time t, denoted Z_t, into a forecast of next-day realized variance through a sequence of transformations and nonlinear activation functions to produce a prediction $\hat{f}(Z_t)$ in the final output layer.

Let L represent the number of layers in a given NN architecture and let N_l denote the total number of neurons in layer l. Formally, the general equation for the l-th hidden layer is provided by Equation (26).

$$a_t^{(\theta_{l+1}, b_{l+1})} = g_l \left(\sum_{j=1}^{N_l} \theta_j^{(l)} a_t^{(\theta_l, b_l)} + b^{(l)} \right), \quad 1 \leq l \leq L \tag{26}$$

In this equation, $g_l(\cdot)$ denotes the activation function, $\theta^{(l)}$ is the weight matrix, $b^{(l)}$ is the bias vector serving as an activation threshold, and $a_t^{(\theta_{l+1},b_{l+1})}$ is the output of layer $l+1$. Note that $a_t^{(\theta_l,b_l)}$ serves as the input layer for layer $l+1$. Note that the output layer uses a linear activation function, which is appropriate for a regression problem where the target variable is continuous. The final forecast is constructed by composing the transformation across all layers, see Equation (27), where each layer adds a new level of transformation to the NN.

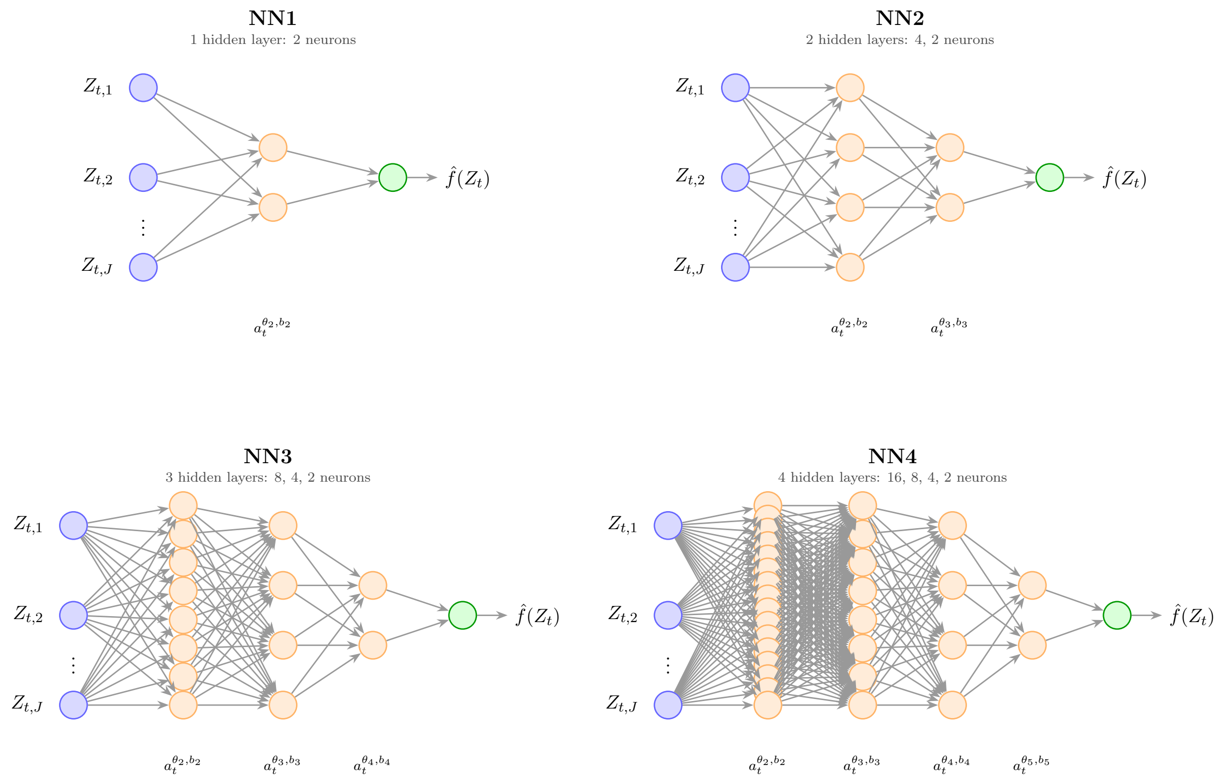
$$\hat{f}(Z_t) = (g_L \circ \cdots \circ g_1)(Z_t) \tag{27}$$

In accordance with Christensen et al. (2023), we adopt the Leaky Rectified Linear Unit (L-ReLU) of Maas et al. (2013) as the activation function for all hidden layers, see Equation (28).

$$\text{L-ReLU}(x) = \begin{cases} cx & \text{if } x < 0 \\ x & \text{otherwise} \end{cases} \tag{28}$$

L-ReLU improves upon the standard ReLU activation function by allowing a small, non-zero slope, c, for negative inputs to prevent neurons from dying in the network. Following Christensen et al. (2023), we set c in Equation (28) equal to 0.01.

Figure 5: Feed-Forward Neural Network Architectures: NN1, NN2, NN3, and NN4



Notes: The figure illustrates the four feed-forward neural network architectures considered in this study, following Christensen et al. (2023) and inspired by the geometric pyramid structure of Masters (1993) and Gu et al. (2020). Each network receives J input features $Z_{t,1}, \ldots, Z_{t,p}$, where $p = 3$ for the $\mathcal{M}_{\text{HAR}}$ feature set and $p = 16$ for the extended $\mathcal{M}_{\text{ALL}}$ feature set. NN_1 is a shallow network with a single hidden layer of 2 neurons; NN_2 has two hidden layers of 4 and 2 neurons; NN_3 has three hidden layers of 8, 4, and 2 neurons; NN_4 has four hidden layers of 16, 8, 4, and 2 neurons. All hidden layers apply a Leaky ReLU activation function with negative slope $c = 0.01$ and are subject to dropout regularization with a keep rate of 0.8. The output layer returns the one-step-ahead forecast $\hat{f}(Z_t)$ via a linear activation function.

4.5.2 Estimation and Regularization

As previously alluded to, all model specifications except for NN architectures themselves are kept fixed. This allows us to attribute any difference in out-of-sample performance solely to the architecture itself as opposed to differing estimation procedures.

First, the neural networks are estimated using the previously defined chronological training, validation and test split, see Table (1). In contrast to models that are re-estimated recursively through time, each neural network is fitted once on the initial training sample and evaluated on the fixed validation and test periods. Importantly, the network is not re-estimated on the union of the training and validation samples prior to forecasting. This fixed-window design follows the empirical implementation in Christensen et al. (2023) and reflects the inherent limitation of the comparatively high computational cost of NN estimation.

Second, both the predictors and the target variable are standardized using moments computed from the initial training sample only. The same transformation is then applied to the validation and test sets. Standardizing the predictors improves numerical stability, while standardizing the target variable facilitates optimization in the training step. Forecasts are subsequently transformed back to the original realized variance scale before proceeding with performance evaluation.

Third, the network parameters are estimated by minimizing the MSE loss function in the standardized target space. Moreover, optimization is performed using the Adam optimizer with a learning rate of 0.001 in accordance with Christensen et al. (2023). Training is conducted using mini-batches of size 64, and the training observations are reshuffled at the beginning och each epoch. A maximum of 500 epochs is allowed and the initial weights are drawn using Xavier (Glorot) normal initialization, while all bias terms are initialized to zero, similarly to the approach of Christensen et al. (2023).

Fourth, regularization is implemented through a combination of dropout and early stopping. Dropout is applied after each hidden layer activation during training according and is used to prevent a NN from becoming to dependent on one or a few single neurons, thereby serving the purpose of reducing overfitting. It works by randomly deactivating neurons in a given layer l during training, see Equation (29) below.

$$\tilde{a}_t^{(\theta_l,b_l)} = d^{(\theta_l,b_l)} \odot a_t^{(\theta_l,b_l)} \quad \text{where} \quad d_i^{(\theta_l,b_l)} \sim \text{Bernoulli}(p) \tag{29}$$

In this equation, $d^{(\theta_l,b_l)}$ is a masked vector performing the drop-out through element-wise multiplication with layer $a_t^{(\theta_l,b_l)}$. We set $p = 0.8$, implying that we keep 80% of the neurons in any given layer l. Dropout is applied only during training and is turned off when generating validation and test forecasts.

To further control overfitting, early stopping is based on the validation set MSE. After each training epoch, the model is evaluated on the full validation set, and training is terminated if the validation loss fails to improve for 100 consecutive epochs. The parameter values from the epoch with the lowest validation MSE are then retained.

Fifth, because NN performance may depend on the random initialization of the weights, each architecture is trained 100 times using a set of different random seeds. The resulting models are ranked according to their validation set MSE. We follow the same approach as Christensen et al. (2023) whereby two forecast variants are retained for each architecture. These are the forecast from the single best-performing network and the forecast obtained by averaging the prediction from the ensemble of the ten best-performing networks. These two variants are denoted with superscripts 1 and 10 respectively in Section (5). This procedure reduces sensitivity to any one particular random initialization while keeping the estimation design comparable across the four architectures.

Lastly, because realized variance is a strictly non-negative variable, an insanity filer is applied to the NN forecasts. Any forecast that falls below the minimum realized variance observed in the pre-test sample, that is , the combined training and validation period, is replaced by that minimum value. This ensures that forecasts remain economically meaningful.

4.6 Out-of-Sample Performance Evaluation

To assess the predictive performance and accuracy of the different forecasting models on each of the two feature sets $\mathcal{M}_{\text{HAR}}$ and $\mathcal{M}_{\text{ALL}}$, we employ a set of different out-of-sample evaluation measures. Following the same methodological approach as Christensen et al. (2023), our primary loss function is the Mean Squared Error (MSE). We supplement this performance measure by computing a pairwise tests of equal predictive accuracy using the framework introduced by Diebold and Mariano (2002). To test the robustness of our findings, we also compute a set of additional loss functions common in the volatility forecasting literature.

4.6.1 Relative Mean Squared Error

Let $\hat{f}(Z_t)$ denote the one-day-ahead forecast of realized variance (RV_{t+1}) provided by one of the forecasting models at time t, based on the feature set Z_t. The feature set Z_t will be either the $\mathcal{M}_{\text{HAR}}$ or the $\mathcal{M}_{\text{ALL}}$ feature set respectively. The MSE of the model is defined in Equation (30).

$$\mathcal{L}_{MSE}(\hat{f}(Z_t)) = \frac{1}{T} \sum_{t \in \text{OOS}} \left(RV_{t+1} - \hat{f}(Z_t) \right)^2 \tag{30}$$

In this equation, the out-of-sample denotes the number of observations, i.e. trading days, in our test sample, and RV_{t+1} is the actual realized variance on trading day $t+1$. Since the MSE performance measure serves as the primary loss function underlying the estimation of all models in our study, it provides a natural and consistent basis for out-of-sample performance evaluation, see Christensen et al. (2023).

To facilitate model comparison, we report a pairwise *relative MSE* score for each model pair (i, j), defined in Equation (31).

$$\text{Relative MSE}_{i,j} = \frac{\text{MSE}_i}{\text{MSE}_j} \tag{31}$$

Here, model j serves as the benchmark. A ratio below 1 indicates that model i outperforms the benchmark model j, while a ratio above 1 implies the opposites. For our results, see Section 5, we report a relative MSE score between each model pair in a more comprehensive format , see Tables (3). For ease of interpretation, we also illustrate the relative MSE och each model relative to the benchmark HAR model of Corsi (2009), see Figures (**??**) since this is one of the most widely used specifications in the realized volatility forecasting literature.

4.6.2 Diebold-Mariano Test

While the relative MSE quantifies the magnitude of the forecasting accuracy of a model pair, it does not provide a format assessment of whether the difference in forecasting accuracy is statistically significant. We therefore complement the relative MSE with the Diebold and Mariano (2002) test for equal predictive accuracy. For a given model pair (i, j), the loss differential series is defined in Equation (32).

$$d_t = \Big(RV_{t+1} - \hat{f}_i(Z_t)\Big)^2 - \Big(RV_{t+1} - \hat{f}_j(Z_t)\Big)^2 \tag{32}$$

In this specification, a negative value of d_t suggests that model i produces a smaller squared error that model j at time t. Moreover, the null hypothesis of equal predictive accuracy is provided in Equation (33) against the one-sided alternative in Equation (34)

$$H_0 : \mathbb{E}[d_t] = 0 \tag{33}$$

$$H_1 : \mathbb{E}[d_t] > 0 \tag{34}$$

The alternative hypothesis H_1 suggests that model j outperforms model i The Diebold and Mariano (2002) test statistic is provided by computing Equation (35).

$$DM = \frac{\bar{d}}{\sqrt{\widehat{\mathrm{Var}}(\bar{d})}} \tag{35}$$

Here, $\bar{d}$ is the sample mean of the loss differential series, and $\widehat{\mathrm{Var}}(\bar{d})$ is a heteroskedasticity and autocorrelation consistent (HAC) variance estimator to account for potential serial correlation in d_t. Under the null hypothesis of equal predictive accuracy, i.e. that none of the models outperforms the other, the test statistic asymptotically follows a standard normal distribution, see Equation (36).

$$DM \xrightarrow{d} \mathcal{N}(0, 1) \tag{36}$$

4.6.3 Additional Loss Functions

While the MSE serves as the primary loss function in our empirical analysis of the results, alternative loss functions may provide complementary insights to evaluate model performance. Therefore, apart from the relative MSE and the pairwise Diebold and Mariano (2002) test, we also report the Quasi Likelihood (QLIKE), Mean Absolute Error (MAE) and the Root Mean Squared Error (RMSE) for each model across both feature sets.

QLIKE is one of the most widely used loss functions in the volatility forecasting literature due to its robustness to measurement error in realized variance, which is an inherently noisy proxy for the latent conditional variance. Unlike MSE, which penalizes squared deviations in levels, QLIKE evaluates forecasts based on the ratio between realized and predictive variance, thereby providing a likelihood-based assessment of model accuracy, see Equation (37). An important feature of QLIKE is its asymmetric penalty structure; it penalizes underestimation of volatility more heavily than overestimation. This is particularly relevant in financial applications, where underpredicting volatility can lead to severe risk management consequences, such as underestimated Value-at-Risk (VaR) or insufficient capital buffers. By placing greater weight on such adverse errors, QLIKE provides a more economically sound assessment of model performance. Moreover, as a proper scoring rule for variance forecasts, QLIKE is minimized when the predictive variance matches the true underlying volatility process. Taken together, these properties make QLIKE a theoretically sound and practically relevant complement to the MSE loss function in the out-of-sample evaluation of model performance.

$$\mathcal{L}_{\text{QLIKE}}(\hat{f}(Z_t)) = \frac{1}{T}\sum_{t\in\text{OOS}}\left[\frac{RV_{t+1}}{\hat{f}(Z_t)} - \log\left(\frac{RV_{t+1}}{\hat{f}(Z_t)}\right) - 1\right] \tag{37}$$

On a different note, the MAE loss function reports the average absolute deviation between the actual one-day-ahead realized variance (RV_{t+1}) and the forecasted one-day-ahead realized variance ($\hat{f}(Z_t)$) for each model. Contrary to the MSE, which squares the forecast errors, the MAE loss function imposes a linear penalty on deviations, thereby making the performance evaluation less sensitive to large forecasting errors. As such, the MAE can be particularly useful in the context of volatility forecasting, where extreme volatility movements, such as those observed during the market turmoil amidst the Covid-19 pandemic, may disproportionally benefit some models. The MAE loss function is presented in Equation (38).

$$\mathcal{L}_{MAE}(\hat{f}(Z_t)) = \frac{1}{T}\sum_{t\in\text{OOS}}\left|RV_{t+1} - \hat{f}(Z_t)\right| \tag{38}$$

The RMSE loss function is closely related to the MSE. By taking the root of the MSE, the RMSE rescales the loss function to the original units of the dependent variable realized variance. Therefore, the RMSE retains the qualities of the quadratic penalization of the MSE while facilitating a more direct economic interpretation of the resulting loss value due to unit consistency. The RMSE is constructed in Equation (39).

$$\mathcal{L}_{RMSE}(\hat{f}(Z_t)) = \sqrt{\frac{1}{T}\sum_{t\in\text{OOS}}\left(RV_{t+1} - \hat{f}(Z_t)\right)^2} \tag{39}$$

5 Empirical Results

In the following sections, we present the empirical findings of our paper. We begin by presenting the performance of each model using only the predictors available in the $\mathcal{M}_{\text{HAR}}$ feature set. We then proceed to compare these results with the results obtained by allowing the more advanced ML models to exploit nonlinearities and complex patterns between the additional variables available in the more extensive $\mathcal{M}_{\text{ALL}}$ feature set.

5.1 Results from $\mathcal{M}_{\text{HAR}}$ Feature Set

In this subsection, we present our empirical findings for all 20 models on the $\mathcal{M}_{\text{HAR}}$ feature set. The out-of-sample test period covers 1,099 trading days spanning the period from 2016-04-06 to 2020-08-06. Following Christensen et al. (2023), the primary loss function is the MSE, reported in pairwise relative form in Table (3). To provide a more comprehensive analysis of the results, we additionally report the corresponding pairwise comparison under the QLIKE loss function in Table (4). Significance markers in both tables refer to a one-sided Diebold and Mariano (2002) test of equal predictive accuracy, where an entry below 1 indicates that the column model produces a lower expected loss than the row model, and the formatting of the values indicate the rejection level of the null hypothesis of equal accuracy in favor of the column model.

5.1.1 Pairwise Relative MSE

Observing the results in Table (3) by columns yields three economically compelling insights. First, HARQ is the single best-performing model on the $\mathcal{M}_{\text{HAR}}$ feature set under the MSE loss function. All other models, including the other benchmark models, regularized linear models, tree-based models, and feed-forward NNs produce a higher MSE value than HARQ. The relative MSE of HARQ compared to HAR is 0.846, with the corresponding Diebold and Mariano (2002) test statistic indicating significance at the 10% level. Against the remaining HAR-family extensions (LogHAR, LevHAR, SHAR), HARQ dominates by roughly $\sim$ 10% to $\sim$ 18% and these gaps are statistically significant in all cases at the 10% level. The dominance of HARQ is consistent with the central insight of Bollerslev et al. (2016) that explicitly correcting for measurement error in lagged realized variance through realized quarticity is particularly valuable in periods that combine extended low-volatility stretches with isolated and extremely volatile trading days, of which our test window contains both. The $\mathcal{M}_{\text{HAR}}$ feature set, restricted to the three lagged realized variance regressors, leaves no further channel through which the remaining models can extract additional information. HARQ therefore wins simply by being the only specification on the $\mathcal{M}_{\text{HAR}}$ feature set that explicitly conditions on second-order properties of realized variance.

Second, the simple HAR benchmark is not unambiguosly dominated by its closest extensions on this single-index series. LogHAR improves on HAR by $\sim$ 6.4% (relative

MSE of 0.936) and is significant at the 10% level, similarly to HARQ. LevHAR improves by a smaller margin (0.951) and the difference is not statistically significant. SHAR is marginally worse than HAR (1.031), again insignificant. This finding aligns with the corresponding result in Christensen et al. (2023), where the basic HAR is shown to be a hard benchmark to systematically beat on the $\mathcal{M}_{\text{HAR}}$ feature set. With only the three lagged realized variance components in the information set, the leverage and signed-semivariance refinements of the LevHAR and SHAR do not deliver consistent improvements at the daily horizon. The HAR-X model collapses mechanically onto HAR in this feature set because no exogenous regressors are available. Therefore, the row and column values for HAR and HAR-X are identical by construction, and we treat it as a single benchmark in the discussion that follows.

Third, the regularized linear models, the three-based models, and the feed-forward neural networks do not improve on HAR under the MSE loss function on the $\mathcal{M}_{\text{HAR}}$ feature set. None of the alternatives in this group rejects the null of equal predictive accuracy in favor of them relative to the benchmark HAR model. Reading the HAR row of Table (3) by columns, RR, LA, and EN all return relative MSE scores above unity (1.028, 1.039, and 1.039 respectively) and the corresponding Diebold and Mariano (2002) test is insignificant. On the contrary, BG, RF, and GB return relative MSE entries below one (0.941, 0.916, and 0.973 respectively), but again none of these models are statistically distinguishable from HAR at conventional significance levels. The same holds across the entire family of feed-forward neural networks. The relative MSE entries of every NN architecture and ensemble specification against HAR cluster tightly around one., ranging from 0.966 (NN_1^1) to 1.045 (NN_4^{10}), with no Diebold and Mariano (2002) rejection in the benchmark HAR model's favor. If anything, reading the shrinkage corresponding columns from inside the LA and EN rows reveals the opposite of an improvement. HAR is significantly better than both LA and EN at the 5% level, and LogHAR is significantly better than both LA and EN at the 1% level. Disciplined linear shrinkage on three predictors therefore does not match HAR or its closest extensions in our European single index setting. We further observe that the LA and EN rows are essentially identical to three decimal points, a pattern that we have verified to arise from cross-validation selecting an effective $_1$-mixing parameter equal to 1 in a substantial share of windows. The consequence of this behavior is that EN reduces to LA on the $\mathcal{M}_{\text{HAR}}$ feature set in our study. The NNs, evaluated under MSE, does not deliver the systematic improvements that Christensen et al. (2023) document for the cross-section of DJIA constituents. This finding is consistent with the interpretation that on a three predictor information set, the marginal value of capturing nonlinearities and complex pattern is small, and that without additional macroeconomic predictors there is little room for deeper NN architectures to learn. The pattern is also internally consistent across architectures. The top-10 ensemble produce relative MSE values close to or marginally worse than the best-seed counterparts, and the depth provides no consistent benefit. We return to the role of the underlying feature set in the discussion in Section (6).

5.1.2 Pairwise Relative QLIKE

Re-evaluating the same forecasts under the QLIKE loss function in Table (4) delivers a markedly sharper picture, both in terms of the magnitude of pairwise differences and in their statistical significance. Three observations stand out.

First, the headline ranking from MSE is preserved under QLIKE in its broad outline, but the gradations are much sharper. HARQ remains a top-tier model under QLIKE, with a relative QLIKE score of 0.792 against HAR, constituting a $\sim 20\%$ improvement that is now significant at the 1% level, compared to the 10% level under the MSE loss function. The other benchmark HAR extensions that materially improves on HAR is LogHAR with a relative QLIKE value of 0.834, again significant at the 1% level. Notably, LevHAR's QLIKE performance reverses sign relative to its MSE result. While its relative MSE score compared to HAR was 0.951 and statistically indistinguishable from one, its QLIKE relative to HAR is 1.733 with strong rejection in the opposite direction. This indicates that LevHAR produces forecasts whose proportional accuracy is materially poorer than HAR even though its MSE performance is similar. This pattern emerges naturally when a model occasionally underestimates volatility on calmer days because the QLIKE loss function penalizes such underestimations more aggressively than the MSE score.

Second, the QLIKE comparison broadens the set of models that significantly outperform HAR to four. The HAR row of Table (4) now records statistically significant rejections at the 1% level not only for HARQ (relative QLIKE of 0.792) but also for LogHAR (0.834), RF (0.781) and BG (0.797). All remaining entries in the HAR row exceed one and carry on no Diebold and Mariano (2002) test significance. RR, LA, and EN record relative QLIKE entries against HAR of 1.686, 1.765 and 1.761 respectively. GB records 1.403, and the NN family records relative QLIKE entries between 1.189 and 1.640. None of these models beats HAR under QLIKE, and reading the corresponing cells in the opposite direction, we observe Diebold and Mariano (2002) rejections in favor of HAR at the 1% significance mark. Disciplined linear shrinkage, GB trees, and the NN family therefore not only fail to improve on HAR under QLIKE on the $\mathcal{M}_{\text{HAR}}$ but are in fact significantly less accurate than HAR under the QLIKE loss function. The economic intuition is that regularized models produce produce forecasts whose proportional accuracy is systematically worse than HAR's. The two model families that do improve on HAR under QLIKE are the same two sub-classes that already showed signs of competitiveness under MSE, namely the HAR-family extensions HARQ and LogHAR and the bootstrap-aggregated tree ensembles BG and RF. What changes between the MSE and the QLIKE application is the strength of the rejection. HARQ and LogHAR move from the 10% to the 1% level under MSE, and BG and RF transition from MSE insignificance into a 1% significance level under QLIKE.

Third, beyond the comparisons against HAR and HARQ, two within-table cross-comparisons are worth noting. The LA and EN rows are again essentially identical, confirming the conclusion drawn from the MSE table that EN collapses onto LA on the three predictor $\mathcal{M}_{\mathrm{HAR}}$ feature set. Within the NN family, the differentiation across architecture depth and across best-seed versus top-10 ensemble is small relative to the gap between the family as a whole and the HAR-family benchmarks. No NN specification rejects the null of equal QLIKE against HAR, HARQ, BG, or RF.

5.1.3 Cross Loss Function Synthesis

Tables (3) and (4) tell a compelling story. Under the MSE loss function, only HARQ and LogHAR reject the null hypothesis of equal predictive accuracy against the HAR benchmark, and only at a 10% significance level. No regularized linear model, tree-based model, or NN specification improves significantly on HAR. These findings contradict that of Christensen et al. (2023), who find that NNs, albeit only marginally and not significantly, improve upon HAR and find no evidence that HARQ is a top performing model. Under the QLIKE loss function, HARQ remains dominant, and the number of models that significantly outperform HAR expands from two to four. Specifically, HARQ and LogHAR strengthen from 10% to 1%, and BG and RF enter the rejection region at the 1% level as well. The regularized linear models, GB, and the NN family do not gain ground under QLIKE. The corresponding Diebold and Mariano (2002) test statistic in fact reject equal predictive accuracy in favor of HAR at the 1% level for all these specifications.

The shift in the rejection pattern between the two losses does not reflect any asymmetry in test construction. Instead, the shift reflects a well-documented property of the two loss functions for volatility forecasting, formalized in Patton (2011) and Patton and Sheppard (2009). Under the MSE loss function, the variance of the per-period loss differential is concentrated in a small number of extremely volatile tradings days on which most models fail simultaneously, which inflates the standard error of the mean differential and depresses the power of the test. Under QLIKE, that variance is more evenly distributed across the sample, so persistent day-to-day differences in proportional forecast accuracy accumulate into a sharper test statistic.

Above all, the $\mathcal{M}_{\mathrm{HAR}}$ results support three key findings. First, HARQ is the best forecast model on the three-predictor information set under the MSE loss function and remains superior under QLIKE. Second, the QLIKE comparison expands the set of models that significantly outperform HAR from two to four, adding BG and RF to HARQ and LogHAR, while the regularized linear models, GB, and NN family are revealed as less significantly less accurate than HAR on the proportional-error metric. Third, the joint reading of the two losses indicates that the $\mathcal{M}_{\mathrm{HAR}}$ feature set leaves little room for non-HAR specifications to overturn the leading benchmark at the one-step-ahead daily horizon forecast, motivating the extension to the more extensive $\mathcal{M}_{\mathrm{ALL}}$ feature set, incorporating macroeconomic variables, in the next section.

Table 3: Pairwise Relative MSE and Diebold-Mariano Tests on $\mathcal{M}_{\text{HAR}}$ Feature Set

	Benchmark Models						Regularized Models			Tree-Based Models			Neural Networks							
	HAR	**HAR-X**	**LogHAR**	**LevHAR**	**SHAR**	**HARQ**	**RR**	**LA**	**EN**	**BG**	**RF**	**GB**	**NN_1^1**	**NN_1^{10}**	**NN_2^1**	**NN_2^{10}**	**NN_3^1**	**NN_3^{10}**	**NN_4^1**	**NN_4^{10}**
HAR	–	1.000	**0.936**	0.951	1.031	**0.846**	1.028	1.039	1.039	0.941	0.916	0.973	0.966	1.009	0.994	0.997	1.005	1.003	1.006	1.045
HAR-X	1.000	–	**0.936**	0.951	1.031	**0.846**	1.028	1.039	1.039	0.941	0.916	0.973	0.966	1.009	0.994	0.997	1.005	1.003	1.006	1.045
LogHAR	1.068	1.068	–	1.016	1.101	**0.904**	1.098	1.110	1.110	1.005	0.979	1.040	1.032	1.078	1.062	1.065	1.074	1.072	1.075	1.117
LevHAR	1.052	1.052	0.984	–	1.084	**0.890**	1.081	1.093	1.093	0.989	0.964	1.024	1.016	1.061	1.046	1.048	1.057	1.055	1.058	1.099
SHAR	0.970	0.970	0.908	0.922	–	**0.821**	0.997	1.008	1.008	0.913	0.889	0.944	0.937	0.979	0.965	0.967	0.975	0.973	0.976	1.014
HARQ	1.182	1.182	1.106	1.124	1.218	–	1.215	1.228	1.228	1.112	1.083	1.150	1.141	1.192	1.175	1.178	1.188	1.186	1.189	1.235
RR	0.973	0.973	**[0.911]**	0.925	1.003	**(0.823)**	–	1.011	1.011	0.915	**0.892**	0.947	**(0.940)**	0.982	0.967	0.970	0.978	0.976	0.979	1.017
LA	**(0.962)**	**(0.962)**	**[0.901]**	0.915	0.992	**(0.814)**	**(0.989)**	–	1.000	0.905	**0.882**	**0.937**	**(0.929)**	0.971	0.957	0.959	0.967	0.966	0.968	1.006
EN	**(0.963)**	**(0.963)**	**[0.901]**	0.915	0.992	**(0.815)**	**(0.989)**	1.000	–	0.905	**0.882**	**0.937**	**(0.930)**	0.971	0.957	0.960	0.968	0.966	0.969	1.006
BG	1.063	1.063	0.995	1.011	1.096	**0.900**	1.093	1.105	1.104	–	0.974	1.035	1.027	1.072	1.057	1.060	1.069	1.067	1.070	1.111
RF	1.091	1.091	1.022	1.038	1.125	**0.924**	1.122	1.134	1.134	1.027	–	1.062	1.054	1.101	1.085	1.088	1.097	1.095	1.098	1.141
GB	1.027	1.027	**0.962**	0.977	1.059	**(0.869)**	1.056	1.068	1.067	0.966	**(0.941)**	–	0.992	1.036	1.022	1.024	1.033	1.031	1.034	1.074
NN_1^1	1.035	1.035	0.969	0.985	1.067	**0.876**	1.064	1.076	1.076	0.974	0.949	1.008	–	1.044	1.029	1.032	1.041	1.039	1.042	1.082
NN_1^{10}	0.991	0.991	0.928	0.943	1.022	**0.839**	1.019	1.030	1.030	0.932	0.908	0.965	**0.957**	–	0.986	0.988	0.997	0.995	0.997	1.036
NN_2^1	1.006	1.006	0.941	0.956	1.037	0.851	1.034	1.045	1.045	0.946	0.922	0.979	0.971	1.015	–	1.003	1.011	1.009	1.012	1.051
NN_2^{10}	1.003	1.003	0.939	0.954	1.034	**0.849**	1.031	1.043	1.042	0.944	0.919	0.976	0.969	1.012	0.997	–	1.008	1.007	1.009	1.049
NN_3^1	0.995	0.995	0.931	0.946	1.025	0.842	1.022	1.034	1.033	0.936	0.911	0.968	0.961	1.003	0.989	0.992	–	0.998	1.001	1.040
NN_3^{10}	0.997	0.997	0.933	0.948	1.027	**0.843**	1.024	1.036	1.035	0.937	0.913	0.970	0.963	1.005	**0.991**	0.993	1.002	–	1.003	1.042
NN_4^1	0.994	0.994	0.930	0.945	1.024	0.841	1.021	1.033	1.032	0.935	0.911	0.967	0.960	1.003	0.988	0.991	0.999	0.997	–	1.039
NN_4^{10}	0.957	0.957	0.896	0.910	0.986	0.810	0.983	0.994	0.994	0.900	0.877	0.931	0.924	0.965	**0.951**	0.954	0.962	0.960	**0.963**	–

Notes: Each entry (i, j) reports the out-of-sample MSE of row model i relative to column model j, computed over the test period 2016-04-21 to 2020-08-06 (1,099 observations). A value below unity indicates that column model j outperforms row model i. Significant values denote rejection of equal predictive accuracy in favor of the column model based on one-sided Diebold and Mariano (2002) tests with Newey-West standard errors (5 lags). The notation for significance level is represented as follows; **number** denotes $p < 0.10$, **(number)** denotes $p < 0.05$, and **[number]** denotes $p < 0.01$.

Table 4: Pairwise Relative QLIKE and Diebold-Mariano Tests on $\mathcal{M}_{\text{HAR}}$ Feature Set

	Benchmark Models						Regularized Models			Tree-Based Models			Neural Networks							
	HAR	**HAR-X**	**LogHAR**	**LevHAR**	**SHAR**	**HARQ**	**RR**	**LA**	**EN**	**BG**	**RF**	**GB**	**NN$_1^1$**	**NN$_1^{10}$**	**NN$_2^1$**	**NN$_2^{10}$**	**NN$_3^1$**	**NN$_3^{10}$**	**NN$_4^1$**	**NN$_4^{10}$**
HAR	–	1.000	**[0.834]**	1.733	1.013	**[0.792]**	1.686	1.765	1.761	**[0.797]**	**[0.781]**	1.403	1.394	1.227	1.222	1.377	1.640	1.448	1.189	1.486
HAR-X	1.000	–	**[0.834]**	1.733	1.013	**[0.792]**	1.686	1.765	1.761	**[0.797]**	**[0.781]**	1.403	1.394	1.227	1.222	1.377	1.640	1.448	1.189	1.486
LogHAR	1.199	1.199	–	2.077	1.215	0.950	2.021	2.116	2.111	0.955	0.937	1.682	1.671	1.470	1.465	1.651	1.966	1.736	1.425	1.782
LevHAR	**[0.577]**	**[0.577]**	**[0.481]**	–	**[0.585]**	**[0.457]**	0.973	1.019	1.017	**[0.460]**	**[0.451]**	**0.810**	**0.805**	**(0.708)**	**(0.705)**	**0.795**	0.947	0.836	**[0.686]**	0.858
SHAR	0.987	0.987	**[0.823]**	1.710	–	**[0.782]**	1.664	1.742	1.738	**[0.786]**	**[0.771]**	1.385	1.376	1.210	1.206	1.359	1.618	1.429	1.173	1.467
HARQ	1.262	1.262	1.053	2.187	1.279	–	2.128	2.228	2.223	1.006	0.986	1.771	1.759	1.548	1.542	1.738	2.070	1.828	1.501	1.876
RR	**[0.593]**	**[0.593]**	**[0.495]**	1.027	**[0.601]**	**[0.470]**	–	1.047	1.044	**[0.473]**	**[0.463]**	**[0.832]**	**[0.827]**	**[0.727]**	**[0.725]**	**[0.817]**	0.972	**[0.859]**	**[0.705]**	**[0.881]**
LA	**[0.567]**	**[0.567]**	**[0.473]**	0.982	**[0.574]**	**[0.449]**	**[0.955]**	–	0.998	**[0.451]**	**[0.443]**	**[0.795]**	**[0.790]**	**[0.695]**	**[0.692]**	**[0.780]**	**(0.929)**	**[0.821]**	**[0.674]**	**[0.842]**
EN	**[0.568]**	**[0.568]**	**[0.474]**	0.984	**[0.575]**	**[0.450]**	**[0.957]**	1.002	–	**[0.452]**	**[0.444]**	**[0.797]**	**[0.792]**	**[0.696]**	**[0.694]**	**[0.782]**	**(0.931)**	**[0.822]**	**[0.675]**	**[0.844]**
BG	1.255	1.255	1.047	2.174	1.272	0.994	2.116	2.215	2.210	–	0.980	1.761	1.750	1.539	1.533	1.728	2.058	1.818	1.492	1.865
RF	1.280	1.280	1.068	2.217	1.297	1.014	2.158	2.259	2.254	1.020	–	1.796	1.784	1.570	1.564	1.763	2.099	1.854	1.522	1.902
GB	**[0.713]**	**[0.713]**	**[0.595]**	1.235	**[0.722]**	**[0.565]**	1.202	1.258	1.255	**[0.568]**	**[0.557]**	–	0.994	**[0.874]**	**[0.871]**	0.981	1.169	1.032	**[0.847]**	1.059
NN$_1^1$	**[0.717]**	**[0.717]**	**[0.598]**	1.243	**[0.727]**	**[0.568]**	1.210	1.266	1.263	**[0.572]**	**[0.560]**	1.006	–	**[0.880]**	**[0.876]**	0.988	1.176	1.039	**[0.853]**	1.066
NN$_1^{10}$	**[0.815]**	**[0.815]**	**[0.680]**	1.413	**[0.826]**	**[0.646]**	1.375	1.439	1.436	**[0.650]**	**[0.637]**	1.144	1.137	–	0.996	1.123	1.337	1.181	**(0.969)**	1.212
NN$_2^1$	**[0.818]**	**[0.818]**	**[0.683]**	1.418	**[0.829]**	**[0.648]**	1.380	1.445	1.441	**[0.652]**	**[0.639]**	1.148	1.141	1.004	–	1.127	1.342	1.185	**[0.973]**	1.216
NN$_2^{10}$	**[0.726]**	**[0.726]**	**[0.606]**	1.258	**[0.736]**	**[0.575]**	1.224	1.282	1.279	**[0.579]**	**[0.567]**	1.019	1.012	**[0.891]**	**[0.887]**	–	1.191	1.052	**[0.863]**	1.079
NN$_3^1$	**[0.610]**	**[0.610]**	**[0.509]**	1.057	**[0.618]**	**[0.483]**	1.028	1.076	1.074	**[0.486]**	**[0.476]**	**[0.856]**	**[0.850]**	**[0.748]**	**[0.745]**	**[0.840]**	–	**[0.883]**	**[0.725]**	**[0.906]**
NN$_3^{10}$	**[0.690]**	**[0.690]**	**[0.576]**	1.196	**[0.700]**	**[0.547]**	1.164	1.219	1.216	**[0.550]**	**[0.539]**	0.969	**(0.963)**	**[0.847]**	**[0.844]**	**[0.951]**	1.132	–	**[0.821]**	1.026
NN$_4^1$	**[0.841]**	**[0.841]**	**[0.702]**	1.457	**[0.852]**	**[0.666]**	1.418	1.485	1.481	**[0.670]**	**[0.657]**	1.180	1.173	1.032	1.028	1.158	1.379	1.218	–	1.250
NN$_4^{10}$	**[0.673]**	**[0.673]**	**[0.561]**	1.166	**[0.682]**	**[0.533]**	1.135	1.188	1.185	**[0.536]**	**[0.526]**	0.944	**[0.938]**	**[0.825]**	**[0.822]**	**[0.927]**	1.103	**[0.974]**	**[0.800]**	–

Notes: Each entry (i, j) reports the out-of-sample QLIKE loss of row model i relative to column model j, computed over the test period 2016-04-21 to 2020-08-06 (1,099 observations). A value below unity indicates that column model j outperforms row model i. Significant values denote rejection of equal predictive accuracy in favour of the column model based on one-sided Diebold and Mariano (2002) tests with Newey–West standard errors (5 lags). The notation for significance level is represented as follows; **number** denotes $p < 0.10$, **(number)** denotes $p < 0.05$, and **[number]** denotes $p < 0.01$.

5.2 Results from $\mathcal{M}_{\text{ALL}}$ Feature Set

This section presents the out-of-sample one-day-ahead realized variance forecasting results for all 20 models estimated on the $\mathcal{M}_{\text{ALL}}$ feature set. As in the $\mathcal{M}_{\text{HAR}}$ analysis, the evaluation panel covers 1,099 trading days from 2016-04-21 to 2020-08-06. Note that the five HAR benchmark models HAR, LogHAR, LevHAR, SHAR, and HARQ are unchanged from the $\mathcal{M}_{\text{HAR}}$ setting and are included here as fixed reference forecasts. The HAR-X model, on the other hand, is re-estimated on the extended $\mathcal{M}_{\text{ALL}}$ feature set together with the regularized linear models, tree-based models, and the NN family. Pairwise relative MSE results are reported in Table (5) and the QLIKE counterpart is reported in (6). Both tables retain the conventions established in Subsection (5.1). Relative entries below one indicate that the column model produces a lower expected loss than the row model, and significance levels reflect the one-sided Diebold and Mariano (2002) test result.

5.2.1 Pairwise Relative MSE

Analyzing Table (5) by columns yields three important observations. First, the rejection front against HAR widens substantially relative to the $\mathcal{M}_{\text{HAR}}$ result. The HAR row of Table (5) records statistically significant rejections at the 10% level for HAR-X (relative MSE of 0.929), LogHAR (0.936), HARQ (0.846), LA (0.908), EN (0.908), RF (0.852), NN1^{10} (0.880), NN2^{1} (0.881) and NN2^{10} (0.895). Whereas the corresponding $\mathcal{M}_{\text{HAR}}$ table identified only HARQ and LogHAR as significant improvements on HAR at the 10% significance level, the addition of the marcoeconomic variables in the $\mathcal{M}_{\text{ALL}}$ feature set brings six new model specifications into the rejection region. The largest point-estimate gains relative to HAR come from HARQ (15.4%) and RF (14.8%), followed by the shallow NN ensembles (NN1^{10} at 12.0% and NN_2^1 at 11.9%) and the regularized linear models LA and EN at 9.2% each.

Second, despite the fact that more models now become more competitive relative to HAR under the $\mathcal{M}_{\text{ALL}}$ feature set, no model significantly outperforms HARQ under the MSE loss function. The HARQ row of Table (5) records relative MSE entries between 1.006 (RF) and 1.218 (SHAR), and none of these models carry Diebold and Mariano (2002) test significance. RF, the closest competitor to HARQ, comes within 0.6% of HARQ's MSE level but the test does not reject equal predictive accuracy. The HARQ result therefore supports the $\mathcal{M}_{\text{HAR}}$ conclusion. Even when the information set expands from three lagged realized variants components to the sixteen predictor $\mathcal{M}_{\text{ALL}}$ feature set, the explicit measurement-error correction in HARQ remains the single hardest benchmark to beat under MSE, in line with what Bollerslev et al. (2016) document.

Third, internal patterns within the $\mathcal{M}_{\text{ALL}}$ model specifications are economically informative. The LA and EN columns are again essentially identical, confirming that cross-validation continues to push the EN L_1-mixing parameter onto the LA boundary in our European single index setting. Within the tree-based family, RF dominates BG (0.8522 versus 0.895 against HAR), the GB model underperform both models with a score of 0.922 against HAR. Within the NN family, the strongest specifications are the shallow networks. $NN1^{10}$, $NN2^{1}$ and $NN2^{10}$ enter the rejection region against HAR, while the deeper NN3 and NN_4 specifications cluster slightly below one but are not statistically indistinguishable from HAR. Top-ten ensembles do not systematically outperform best-seed counterparts within architecture depth.

Taken together, the relative MSE results from the $\mathcal{M}_{\text{ALL}}$ feature set differs materially from Table (3). The HAR benchmark is no longer a sharp separator, nine alternative models now produce significantly lower MSE than HAR at the 10% level. The competitive landscape in terms of predictive performance has shifted from a HAR versus HARQ and LogHAR contrast on the three predictor information set toward a wider competitive band in which regularized models, tree-based models, and shallow NNs that now are statistically distinguishable from HAR. The MSE ranking nonetheless preserves HARQ as the leading specification on the EURO STOXX 50 single index series under the MSE loss function.

5.2.2 Pairwise Relative QLIKE

Before we proceed with the analysis of the relative QLIKE results, two preliminary observations are required before reading Table (6). The HAR-X and RR regression columns under $\mathcal{M}_{\text{ALL}}$ contain extreme QLIKE values (relative QLIKE against HAR of 1685.4 and 44.9 respectively). The corresponding rows mirror these magnitudes. We investigate this issue and find that this problem stems from the fact that the HAR-X and RR model produce many negative forecasts during the test period. Since realized variance can not be negative, these values are replaced via the application of the insanity filter with the minimum realized variance observed during the training window. Since these values are very small, and this procedure is repeated frequently due to negative forecasts, it causes the QLIKE and relative QLIKE score to effectively explode, see e.g. Table (13) in Appendix (4.6). The economic interpretation of the HAR-X and RR regression figures is that, on the $\mathcal{M}_{\text{ALL}}$ feature set, neither model produces forecasts that remain inside the empirical support of realized variance with sufficient regularity to be evaluated coherently under the QLIKE loss function. With this in mind, three key observations follow.

First, the QLIKE-based rejection front against HAR contains five models. The HAR row of Table (6) records statistically significant rejections at the 1% level for LogHAR (relative QLIKE of 0.834), HARQ (0.792), BG (0.739), RF (0.732), and GB (0.887). The LogHAR and HARQ entries are mechanically identical to Table (4) by construction. The novel additions relative to $\mathcal{M}_{\text{HAR}}$ are the strengthening of BG and RF from $\mathcal{M}_{\text{HAR}}$ relative QLIKE values of 0.797 and 0.781 to $\mathcal{M}_{\text{ALL}}$ entries of 0.739 and 0.732. Furthermore, under $\mathcal{M}_{\text{HAR}}$, GB recorded a relative QLIKE score of 1.403 against HAR with no rejection, whereas under $\mathcal{M}_{\text{ALL}}$ it improves to 0.887 with rejection at the 1% level. The macroeconomic covariates therefore deliver tangible QLIKE gains for all tree-based models, with GB in particular benefiting most from the broader information set.

Second, the QLIKE comparison narrows the $\mathcal{M}_{\text{ALL}}$ leadership ordering at the top of the model ranking. The HARQ row of Table (6) records significant rejections at the 5% level for BG (relative QLIKE of 0.932) and at the 1% level for RF (0.923). This is a substantive transition relative to both the $\mathcal{M}_{\text{HAR}}$ QLIKE result and the $\mathcal{M}_{\text{ALL}}$ MSE result. Under $\mathcal{M}_{\text{HAR}}$, no model significantly outperformed HARQ on either loss function. Under $\mathcal{M}_{\text{ALL}}$ MSE, the RF model came within 0.6% of HARQ but failed to reject. Under $\mathcal{M}_{\text{ALL}}$ QLIKE, however, the same RF specification rejects equal predictive accuracy against HARQ at the 1%, and BG rejects at the 5% level. The proportional error metric thus identifies the bootstrap-aggregated tree-based ensembles as the only specifications in our $\mathcal{M}_{\text{ALL}}$ model space capable of statistically dominating the HARQ benchmark in our EURO STOXX 50 single index setting.

Third, the regularized linear models and NN family do not share the same QLIKE gains documented for the tree-based ensembles. Reading the HAR row in Table (6), LA en EN record relative QLIKE entries of 1.090 and 1.093 against HAR with no rejection, despite their statistically significant 9.2% MSE improvement on $\mathcal{M}_{\text{ALL}}$. Every NN specification records a relative Q-LIKE entry above one against HAR, ranging from 1.103 (NN1^1) to 1.661 (NN4^{10}), again with no rejection. The reverse direction tests inside the same table reject equal predictive accuracy in HAR's favor at the 1% level for every regularized linear and NN specification. The MSE-based gains delivered by these specifications on $\mathcal{M}_{\text{ALL}}$ therefore do not translate into improvements on the proportional error metric. Under QLIKE, HAR significantly dominates the regularized linear models and the entire NN family.

5.2.3 Cross Loss Function Synthesis

Tables (5) and (6) jointly support a sharper characterization of the $\mathcal{M}_{\text{ALL}}$ feature set than either table alone. Under the MSE loss function, the addition of the macroeconomic variables expands the set of models that significantly outperform HAR from two specifications under $\mathcal{M}_{\text{HAR}}$ (HARQ and LogHAR) to nine specifications under MALL (HAR-X, LogHAR, HARQ, LA, EN, RF, NN_1^{10}, NN_2^1 and NN_2^{10}). HARQ remains undominated by any competing model under the MSE function on $\mathcal{M}_{\text{ALL}}$. Under QLIKE, the rejection front against HAR contains LogHAR, HARQ, BG, RF, and GB, with the latter three reflecting genuine $\mathcal{M}_{\text{ALL}}$ driven improvements relative to the $\mathcal{M}_{\text{HAR}}$ analysis. More consequentially, the QLIKE comparison identifies BG and RF as significant improvements over HARQ at the 5% and 1% level respectively, marking the first instance in the empirical analysis in which any specification statistically dominates HARQ. The NN family and the regularized linear models LA and EN deliver MSE gains over HAR that are not supported under QLIKE. Under the proportional error metric, these classes are significantly less accurate than HAR, indicating that their MSE gains are driven by improved fit on extremely volatile days rather than by uniform proportional accuracy across the test window. The contrast between the MSE and QLIKE rankings is the sharpest for the NNs, whose MSE-best specification (NN1^{10}, NN2^{1}, NN2^{10}) are also among the specifications most decisively dominated by HAR under the QLIKE loss function.

Reflecting upon these findings, the $\mathcal{M}_{\text{ALL}}$ results support three main conclusions. First, the macrofinancial covariates deliver a measureable expansion of the set of models that significantly outperform HAR under the MSE loss function, but they do not overturn HARQ as the leading benchmark on this metric. This finding suggests that all ML models improve their MSE score relative to under $\mathcal{M}_{\text{HAR}}$, suggesting that they are able to exploit the additional information incorporated in the more extensive $\mathcal{M}_{\text{ALL}}$ feature set. Second, the QLIKE comparison identifies the bootstrapped-aggregated tree-based ensembles, BG and RF, as the only specifications in the $\mathcal{M}_{\text{ALL}}$ model space capable of statistically improving on HARQ on the proportional-error metric, with RF doing so at a 1% significance level. Third, the joint reading of the MSE and QLIKE indicates that the MSE-based improvements documented for the regularized linear models and NN family on $\mathcal{M}_{\text{ALL}}$ do not extend to QLIKE, suggesting that the channels through which these specifications absorb the marcoeconomic information differ qualitatively from those exploited by the three-based models. The economic interpretation of these patterns and their implications for the use of ML forecasts across different volatility regimes are discussed further in Section (6).

Table 5: Pairwise Relative MSE and Diebold-Mariano Tests on $\mathcal{M}_{\text{ALL}}$ Feature Set

	Benchmark Models						Regularized Models			Tree-Based Models			Neural Networks							
	HAR	**HAR-X**	**LogHAR**	**LevHAR**	**SHAR**	**HARQ**	**RR**	**LA**	**EN**	**BG**	**RF**	**GB**	**NN$_1^1$**	**NN$_1^{10}$**	**NN$_2^1$**	**NN$_2^{10}$**	**NN$_3^1$**	**NN$_3^{10}$**	**NN$_4^1$**	**NN$_4^{10}$**
HAR	–	**0.929**	**0.936**	0.951	1.031	**0.846**	0.922	**0.908**	**0.908**	0.895	**0.852**	0.922	0.944	**0.880**	**0.881**	**0.895**	0.924	0.943	0.961	0.946
HAR-X	1.077	–	1.008	1.024	1.110	**0.911**	0.993	0.978	0.978	0.964	0.917	0.992	1.017	**0.947**	0.949	0.963	0.995	1.015	1.035	1.018
LogHAR	1.068	0.992	–	1.016	1.101	**0.904**	0.985	0.970	0.971	0.956	0.910	0.985	1.009	**0.940**	**0.941**	**0.956**	0.987	1.008	1.027	1.010
LevHAR	1.052	0.977	0.984	–	1.084	**0.890**	0.970	0.955	0.955	0.941	**0.896**	0.969	0.993	0.925	0.927	0.941	0.972	0.992	1.011	0.994
SHAR	0.970	**0.901**	0.908	0.922	–	**0.821**	0.894	**0.881**	**0.881**	0.868	**0.826**	0.894	0.916	**0.853**	**0.855**	**0.868**	0.896	0.915	0.932	0.917
HARQ	1.182	1.098	1.106	1.124	1.218	–	1.089	1.073	1.074	1.058	1.006	1.089	1.116	1.039	1.041	1.057	1.092	1.114	1.136	1.117
RR	1.085	1.008	1.015	1.031	1.118	**0.918**	–	**0.985**	**0.985**	0.971	0.924	1.000	1.024	**0.954**	**0.956**	**(0.971)**	1.002	1.023	1.042	1.026
LA	1.101	1.023	1.030	1.047	1.135	0.932	1.015	–	1.000	0.985	0.938	1.015	1.040	0.968	0.970	0.985	1.017	1.038	1.058	1.041
EN	1.101	1.023	1.030	1.047	1.135	0.931	1.015	1.000	–	0.985	0.937	1.014	1.040	0.968	0.970	0.985	1.017	1.038	1.058	1.041
BG	1.117	1.038	1.046	1.062	1.152	0.945	1.030	1.015	1.015	–	**0.951**	1.030	1.055	0.983	0.984	1.000	1.032	1.054	1.074	1.056
RF	1.174	1.091	1.099	1.117	1.210	0.994	1.083	1.067	1.067	1.051	–	1.082	1.109	1.033	1.035	1.051	1.085	1.107	1.129	1.110
GB	1.085	1.008	1.016	1.032	1.118	0.918	1.000	0.986	0.986	0.971	0.924	–	1.025	0.954	0.956	0.971	1.002	1.023	1.043	1.026
NN$_1^1$	1.059	0.984	0.991	1.007	1.092	**0.896**	0.976	0.962	0.962	0.948	**0.902**	0.976	–	**0.931**	**0.933**	**0.948**	0.978	0.999	1.018	1.001
NN$_1^{10}$	1.137	1.056	1.064	1.081	1.172	0.962	1.048	1.033	1.033	1.018	0.968	1.048	1.074	–	1.002	1.017	1.050	1.072	1.093	1.075
NN$_2^1$	1.135	1.054	1.062	1.079	1.170	0.960	1.046	1.031	1.031	1.016	0.966	1.046	1.072	0.998	–	1.016	1.048	1.070	1.091	1.073
NN$_2^{10}$	1.118	1.038	1.046	1.063	1.152	0.946	1.030	1.015	1.015	1.000	0.952	1.030	1.055	0.983	0.985	–	1.032	1.054	1.074	1.057
NN$_3^1$	1.082	1.005	1.013	1.029	1.116	0.916	0.998	0.983	0.983	0.969	0.922	0.998	1.022	0.952	0.954	0.969	–	1.021	1.040	1.024
NN$_3^{10}$	1.060	0.985	0.993	1.008	1.093	0.897	0.978	0.963	0.963	0.949	0.903	0.977	1.001	0.933	0.934	0.949	**(0.980)**	–	1.019	1.003
NN$_4^1$	1.041	0.967	0.974	0.989	1.073	**0.880**	0.959	**0.945**	**0.945**	0.931	0.886	0.959	0.983	0.915	**0.917**	**0.931**	**(0.961)**	**(0.981)**	–	0.984
NN$_4^{10}$	1.058	0.982	0.990	1.006	1.090	**0.895**	0.975	**0.961**	**0.961**	0.947	0.901	0.975	0.999	**0.930**	**0.932**	**0.946**	**[0.977]**	0.997	1.016	–

Notes: Each entry (i, j) reports the out-of-sample MSE of row model i relative to column model j, computed over the test period 2016-04-21 to 2020-08-06 (1,099 observations). A value below unity indicates that column model j outperforms row model i. Significant values denote rejection of equal predictive accuracy in favor of the column model based on one-sided Diebold and Mariano (2002) tests with Newey-West standard errors (5 lags). The notation for significance level is represented as follows; **number** denotes $p < 0.10$, **(number)** denotes $p < 0.05$, and **[number]** denotes $p < 0.01$.

Table 6: Pairwise Relative QLIKE and Diebold-Mariano Tests on $\mathcal{M}_{\text{ALL}}$ Feature Set

	Benchmark Models						Regularized Models			Tree-Based Models			Neural Networks							
	HAR	**HAR-X**	**LogHAR**	**LevHAR**	**SHAR**	**HARQ**	**RR**	**LA**	**EN**	**BG**	**RF**	**GB**	$\mathbf{NN}_1^1$	$\mathbf{NN}_1^{10}$	$\mathbf{NN}_2^1$	$\mathbf{NN}_2^{10}$	$\mathbf{NN}_3^1$	$\mathbf{NN}_3^{10}$	$\mathbf{NN}_4^1$	$\mathbf{NN}_4^{10}$
HAR	–	1685.437	**[0.834]**	1.733	1.013	**[0.792]**	44.909	1.090	1.093	**[0.739]**	**[0.732]**	**[0.887]**	1.103	1.315	1.317	1.174	1.136	1.345	1.559	1.661
HAR-X	**[0.001]**	–	**[0.000]**	**[0.001]**	**[0.001]**	**[0.000]**	**[0.027]**	**[0.001]**	**[0.001]**	**[0.000]**	**[0.000]**	**[0.001]**	**[0.001]**	**[0.001]**	**[0.001]**	**[0.001]**	**[0.001]**	**[0.001]**	**[0.001]**	**[0.001]**
LogHAR	1.199	2020.422	–	2.077	1.215	0.950	53.835	1.307	1.310	**[0.885]**	**[0.877]**	1.063	1.322	1.577	1.579	1.408	1.361	1.612	1.869	1.991
LevHAR	**[0.577]**	972.825	**[0.481]**	–	**[0.585]**	**[0.457]**	25.921	**[0.629]**	**[0.631]**	**[0.426]**	**[0.422]**	**[0.512]**	**[0.636]**	**(0.759)**	**(0.760)**	**[0.678]**	**[0.655]**	**(0.776)**	0.900	0.959
SHAR	0.987	1663.146	**[0.823]**	1.710	–	**[0.782]**	44.315	1.076	1.078	**[0.729]**	**[0.722]**	**[0.875]**	1.088	1.298	1.299	1.159	1.120	1.327	1.538	1.639
HARQ	1.262	2127.090	1.053	2.187	1.279	–	56.678	1.376	1.379	**(0.932)**	**[0.923]**	1.119	1.392	1.660	1.662	1.482	1.433	1.697	1.968	2.096
RR	**[0.022]**	37.530	**[0.019]**	**[0.039]**	**[0.023]**	**[0.018]**	–	**[0.024]**	**[0.024]**	**[0.016]**	**[0.016]**	**[0.020]**	**[0.025]**	**[0.029]**	**[0.029]**	**[0.026]**	**[0.025]**	**[0.030]**	**[0.035]**	**[0.037]**
LA	**(0.917)**	1545.679	**[0.765]**	1.589	**0.929**	**[0.727]**	41.186	–	1.002	**[0.677]**	**[0.671]**	**[0.813]**	1.011	1.206	1.208	1.077	1.041	1.233	1.430	1.523
EN	**(0.915)**	1542.180	**[0.763]**	1.585	**(0.927)**	**[0.725]**	41.092	**[0.998]**	–	**[0.676]**	**[0.669]**	**[0.811]**	1.009	1.203	1.205	1.074	1.039	1.230	1.426	1.520
BG	1.354	2281.905	1.129	2.346	1.372	1.073	60.803	1.476	1.480	–	0.991	1.201	1.493	1.781	1.783	1.590	1.537	1.821	2.111	2.249
RF	1.367	2303.578	1.140	2.368	1.385	1.083	61.380	1.490	1.494	1.009	–	1.212	1.507	1.798	1.800	1.605	1.552	1.838	2.131	2.270
GB	1.128	1900.558	0.941	1.954	1.143	**(0.894)**	50.641	1.230	1.232	**[0.833]**	**[0.825]**	–	1.243	1.483	1.485	1.324	1.280	1.516	1.758	1.873
$\mathbf{NN}_1^1$	**[0.907]**	1528.630	**[0.757]**	1.571	**[0.919]**	**[0.719]**	40.731	0.989	0.991	**[0.670]**	**[0.664]**	**[0.804]**	–	1.193	1.194	1.065	1.030	1.220	1.414	1.507
$\mathbf{NN}_1^{10}$	**[0.760]**	1281.469	**[0.634]**	1.317	**[0.771]**	**[0.602]**	34.146	**[0.829]**	**[0.831]**	**[0.562]**	**[0.556]**	**[0.674]**	**[0.838]**	–	1.001	**[0.893]**	**[0.863]**	1.022	1.185	1.263
$\mathbf{NN}_2^1$	**[0.759]**	1279.914	**[0.633]**	1.316	**[0.770]**	**[0.602]**	34.104	**[0.828]**	**[0.830]**	**[0.561]**	**[0.556]**	**[0.673]**	**[0.837]**	0.999	–	**[0.892]**	**[0.862]**	1.021	1.184	1.261
$\mathbf{NN}_2^{10}$	**[0.852]**	1435.336	**[0.710]**	1.475	**[0.863]**	**[0.675]**	38.245	**(0.929)**	**0.931**	**[0.629]**	**[0.623]**	**[0.755]**	**[0.939]**	1.120	1.121	–	0.967	1.145	1.328	1.415
$\mathbf{NN}_3^1$	**[0.881]**	1484.306	**[0.735]**	1.526	**[0.892]**	**[0.698]**	39.550	**0.960**	0.962	**[0.650]**	**[0.644]**	**[0.781]**	0.971	1.158	1.160	1.034	–	1.184	1.373	1.463
$\mathbf{NN}_3^{10}$	**[0.744]**	1253.408	**[0.620]**	1.288	**[0.754]**	**[0.589]**	33.398	**[0.811]**	**[0.813]**	**[0.549]**	**[0.544]**	**[0.659]**	**[0.820]**	**[0.978]**	**(0.979)**	**[0.873]**	**[0.844]**	–	1.159	1.235
$\mathbf{NN}_4^1$	**[0.641]**	1081.104	**[0.535]**	1.111	**[0.650]**	**[0.508]**	28.807	**[0.699]**	**[0.701]**	**[0.474]**	**[0.469]**	**[0.569]**	**[0.707]**	**[0.844]**	**[0.845]**	**[0.753]**	**[0.728]**	**[0.863]**	–	1.065
$\mathbf{NN}_4^{10}$	**[0.602]**	1014.669	**[0.502]**	1.043	**[0.610]**	**[0.477]**	27.036	**[0.656]**	**[0.658]**	**[0.445]**	**[0.440]**	**[0.534]**	**[0.664]**	**[0.792]**	**[0.793]**	**[0.707]**	**[0.684]**	**[0.810]**	**[0.939]**	–

Notes: Each entry (i, j) reports the out-of-sample QLIKE loss of row model i relative to column model j, computed over the test period 2016-04-21 to 2020-08-06 (1,099 observations). A value below unity indicates that column model j outperforms row model i. Significant values denote rejection of equal predictive accuracy in favour of the column model based on one-sided Diebold and Mariano (2002) tests with Newey–West standard errors (5 lags). The notation for significance level is represented as follows; **number** denotes $p < 0.10$, **(number)** denotes $p < 0.05$, and **[number]** denotes $p < 0.01$.

6 Discussion

6.1 Performance under Different Volatility Regimes

6.2 Variable Importance

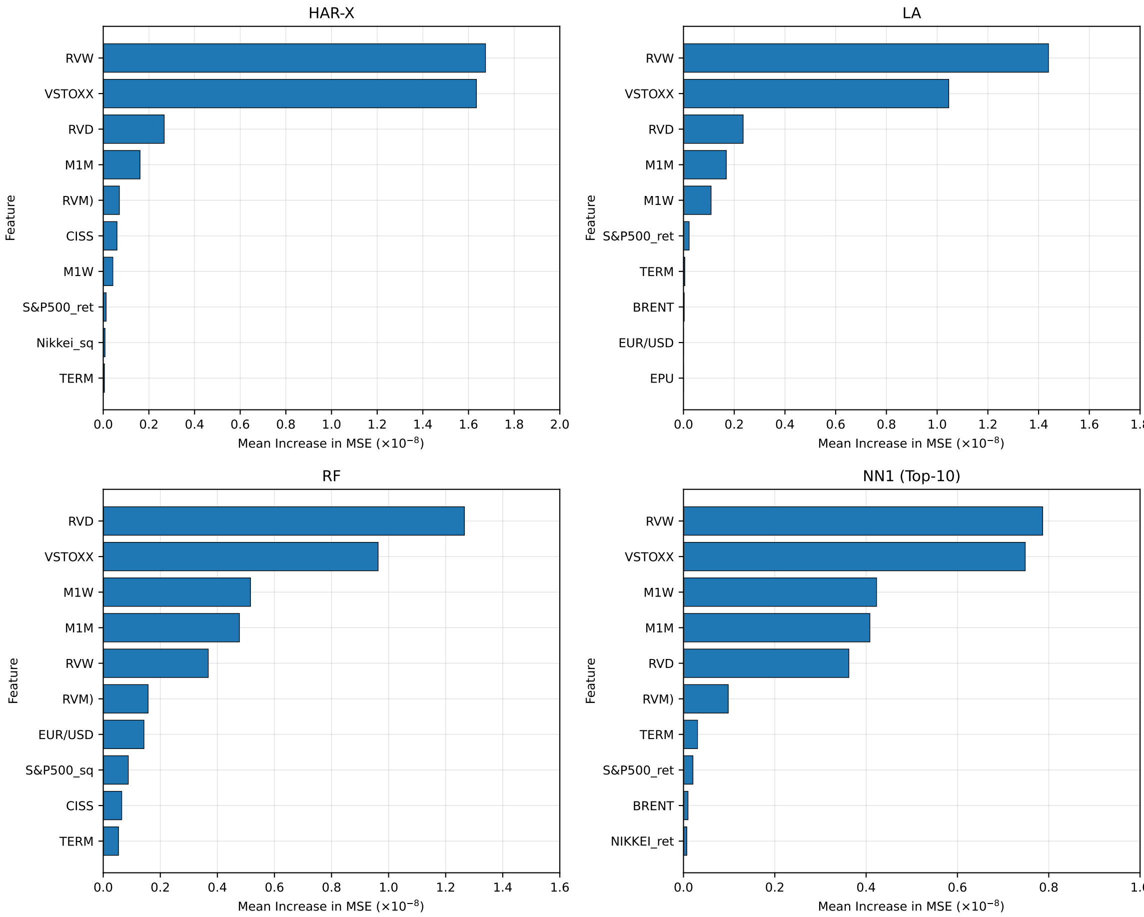
In this subsection, we investigate what variables are most important to drive forecasting performance across all four model categories. We proceed to investigate the best performing model from each model category, in terms of relative MSE over the benchmark HAR model, from Table (5). Note that, albeit HARQ is the best performing benchmark model in this setting, we choose to analyze the variable importance of the exogenous predictors of the HAR-X model instead since this model applies all additional macroeconomic predictors available in the $\mathcal{M}_{\text{ALL}}$ feature set. This approach provides us with a holistic view of what variables drive predictive performance and whether the importance differs between simple linear regression models, regularized linear regression models, tree-based models, and feed-forward NNs respectively.

6.3 Limitations

6.4 Future Research

7 Concluding Remarks

Figure 6: Key Drivers of Realized Volatility Forecasts

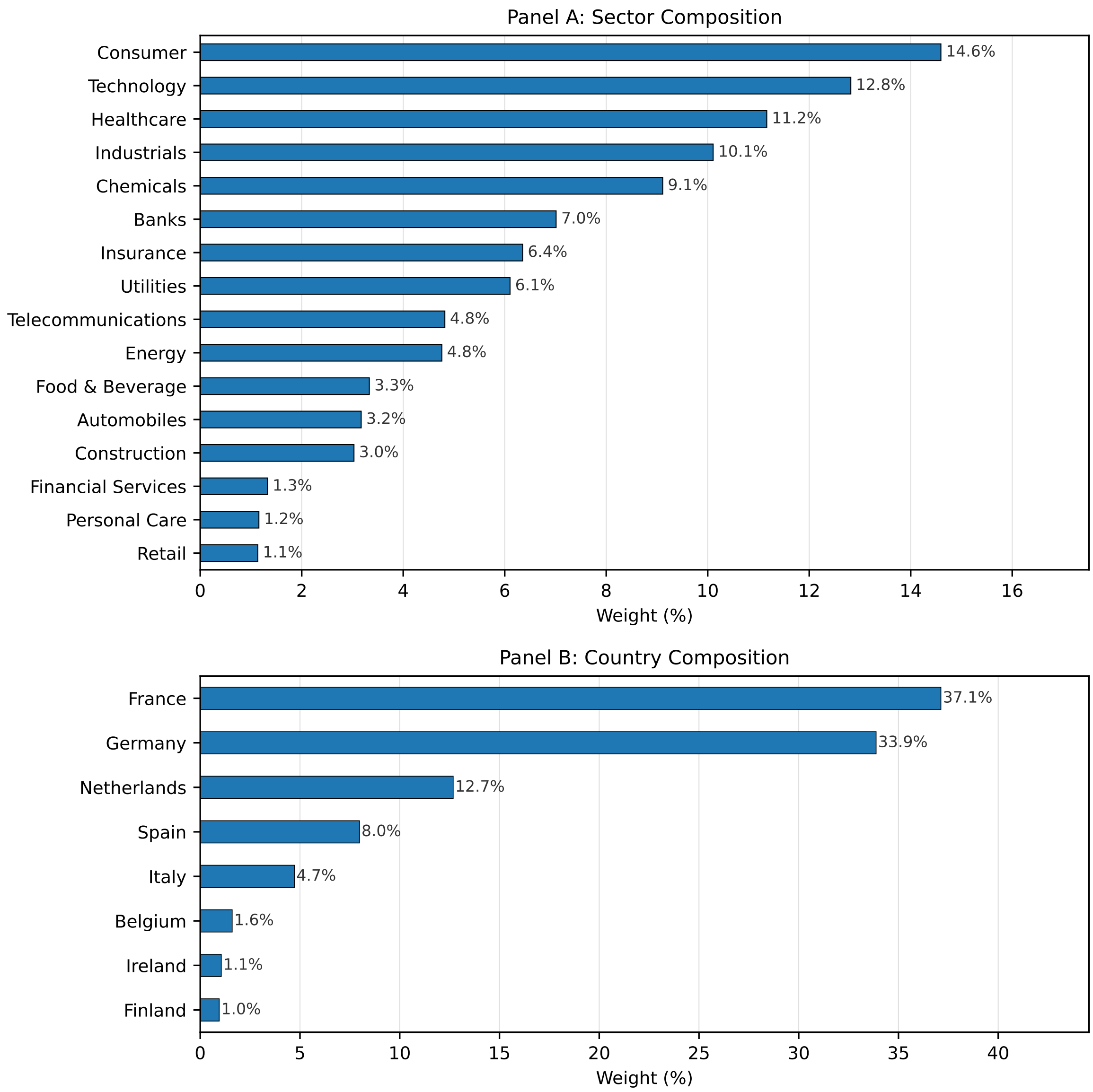


Notes: This figure reports permutation feature importance for the best performing models from each model family. For a given predictor, importance is measured as the average increase in out-of-sample mean squared error (MSE) when the observations of that variable are randomly permuted, thereby corrupting its relationship with the target variable realized variance. All importance measures are computed on the test set and are based on the most recent rolling estimation window. Results are averaged over 30 random permutations, where larger values imply that the model is relying heavily on that feature to forecast realized variance whereas lower values indicate that the variable bears little predictive power. Each panel shows the 10 most important variables.

8 Appendix

8.1 EURO STOXX 50 Composition

Figure 7: EURO STOXX 50 Composition by Sector and Country (2020)



Notes: The figure plots the sector and country composition of the EURO STOXX 50 index as of June 30, 2020. The top panel shows the distribution across industry sectors, while the bottom panel presents the distribution across countries. Please note that this is only a snapshot in time, as composition data is only available on a quarterly basis from 2019 onwards. As such, the weights are not representative of the entire sample period, but provide a general overview of the index, with weights expected to remain relatively stable. Index constituents, sector classifications, and weights are sourced from the STOXX website.

8.2 Volatile Events

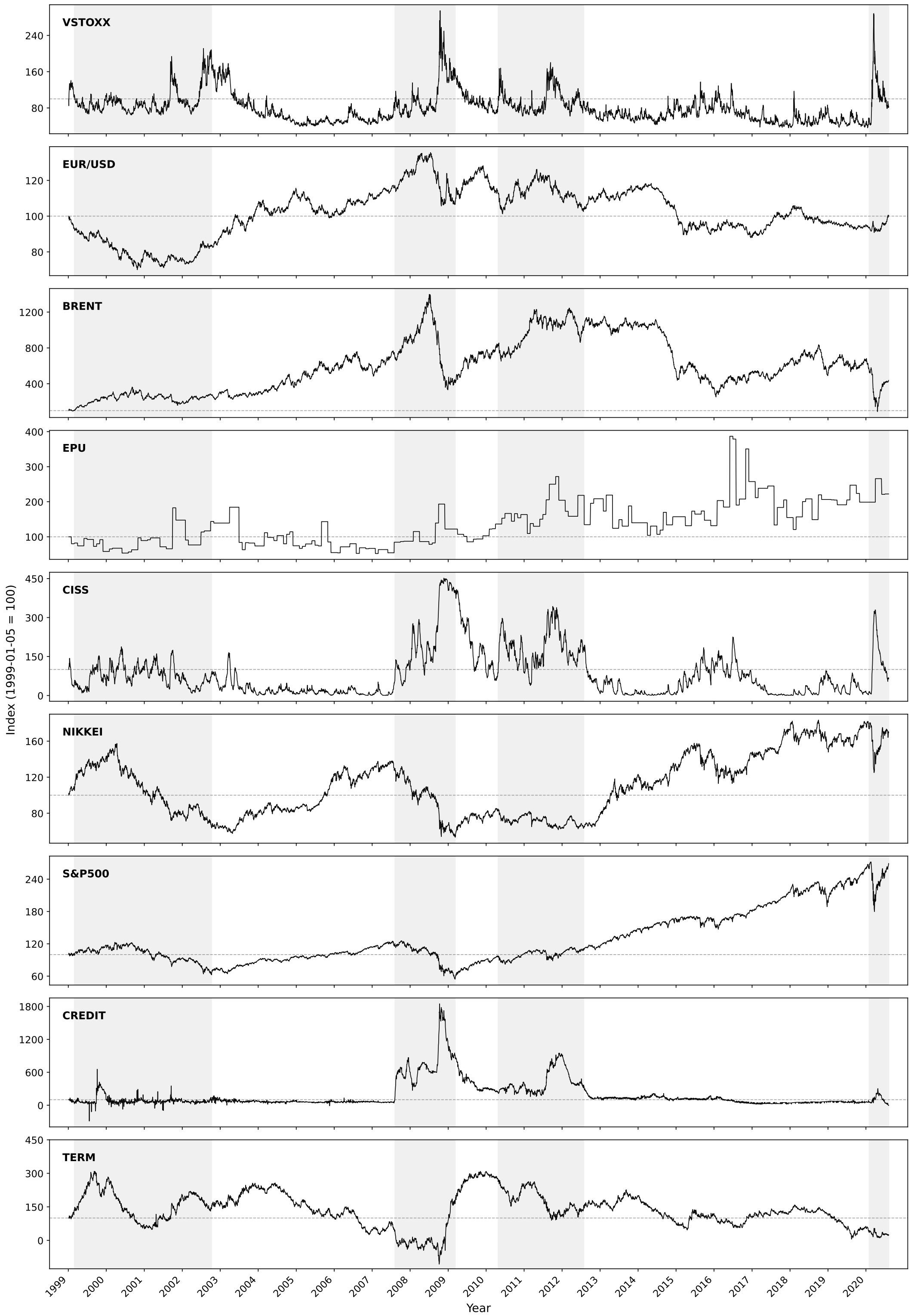
Table 7: Description of Annual Volatile Event Outliers on the EURO STOXX 50

Year	**Date**	**Description of Volatile Event**
1999	Jan. 15	Brazil floated the real, abandoning its dollar peg and triggering devaluation.
2000	Apr. 17	Dot-com bubble burst; the NASDAQ closed its worst week on record.
2001	Sep. 11	Al-Qaeda attacks on the World Trade Center and the Pentagon.
2002	Jul. 25	WorldCom's Chapter 11 filing amid widespread U.S. accounting scandals.
2003	Mar. 17	U.S.-led coalition issued its 48-hour ultimatum ahead of the Iraq invasion.
2004	Mar. 11	Coordinated Al-Qaeda bombings on Madrid's commuter rail network.
2005	Jul. 07	Four coordinated suicide bombings struck London's transport network.
2006	May 18	Sharp correction driven by U.S. inflation and Fed rate hike expectations.
2007	Aug. 17	Fed announced an emergency 50bp discount rate cut as credit markets froze.
2008	Oct. 10	Post-Lehman crisis peak; equities' worst weekly performance since 1929.
2009	Jan. 23	U.K. confirmed its first recession since 1991, deepening recession fears.
2010	May 07	Greece's emergency €110bn IMF-EU bailout failed to contain contagion.
2011	Aug. 11	S&P's first-ever U.S. downgrade and surging Italian and Spanish bond yields.
2012	Aug. 02	Aftermath of Draghi's *"whatever it takes"* pledge (Jul. 26).
2013	Feb. 25	Inconclusive Italian elections produced a hung parliament.
2014	Oct. 16	IMF growth downgrades, falling oil prices, and Ebola fears triggered a sell-off.
2015	Aug. 24	"Black Monday": China's Shanghai Composite plunged over 8%.
2016	Jun. 24	The U.K.'s vote to leave the European Union triggered a record one-day drop.
2017	Apr. 24	First-round French election eased Le Pen fears, sparking a relief rally.
2018	Feb. 06	The *"Volmageddon"* VIX spike collapsed inverse-volatility ETPs.
2019	Aug. 15	U.S.-China trade escalation and the yuan's break past 7 per dollar.
2020	Mar. 16	"Black Thursday" aftermath as COVID-19 lockdowns spread across Europe.

Notes: This table provides an event description for each of the volatile event outliers presented in the boxplot in Figure (3) for the entire sample period from the 5th of January 1999 to the 6th of August 2020.

8.3 Indexed Evolution of the Macroeconomic Predictors

Figure 8: Macroeconomic Variables Panel Included in $\mathcal{M}_{\text{ALL}}$ Feature Set



Notes: This figure presents the indexed values macroeconomic variables included in the $\mathcal{M}_{\text{ALL}}$ feature set. The index for each variable is set at 100 at the start of the sample period, on the 5th of January 1999. The sample period runs from the 5th of January 1999 to the 6th of August 2020. Shaded regions indicate significant and persistent volatility regimes, previously described in Figure (2).

8.4 EURO STOXX 50 Index Constituents

Table 8: An Overview of the EURO STOXX 50 Index Constituents (2020)

Company	**Sector**	**Country**	**Weight (%)**
ASML HLDG	Technology	Netherlands	6.025
SAP	Technology	Germany	5.890
LINDE	Chemicals	Germany	4.498
LVMH MOET HENNESSY	Consumer	France	4.492
SANOFI	Healthcare	France	4.466
TOTAL	Energy	France	3.828
SIEMENS	Industrials	Germany	3.394
ALLIANZ	Insurance	Germany	3.283
L'OREAL	Consumer	France	3.014
UNILEVER NV	Consumer	Netherlands	2.806
BAYER	Healthcare	Germany	2.799
IBERDROLA	Utilities	Spain	2.637
AIR LIQUIDE	Chemicals	France	2.632
ENEL	Utilities	Italy	2.583
SCHNEIDER ELECTRIC	Industrials	France	2.359
DEUTSCHE TELEKOM	Telecommunications	Germany	2.098
BASF	Chemicals	Germany	1.984
VINCI	Construction	France	1.974
ADIDAS	Consumer	Germany	1.863
BNP PARIBAS	Banks	France	1.766
DANONE	Food & Beverage	France	1.730
AXA	Insurance	France	1.667
PHILIPS	Healthcare	Netherlands	1.646
ANHEUSER-BUSCH INBEV	Food & Beverage	Belgium	1.603
AIRBUS	Industrials	France	1.595
BCO SANTANDER	Banks	Spain	1.565
KERING	Consumer	France	1.563
ESSILORLUXOTTICA	Healthcare	France	1.467
MUENCHENER RUECK	Insurance	Germany	1.404
DEUTSCHE POST	Industrials	Germany	1.385
SAFRAN	Industrials	France	1.374
DEUTSCHE BOERSE	Financial Services	Germany	1.325
DAIMLER	Automobiles	Germany	1.314
INTESA SANPAOLO	Banks	Italy	1.204
AHOLD DELHAIZE	Personal Care	Netherlands	1.156
INDUSTRIA DE DISENO TEXTIL SA	Retail	Spain	1.134
VOLKSWAGEN PREF	Automobiles	Germany	1.070
CRH	Construction	Ireland	1.056
ING GRP	Banks	Netherlands	1.047
NOKIA	Telecommunications	Finland	0.952
ENI	Energy	Italy	0.934
ORANGE	Telecommunications	France	0.913
AMADEUS IT GROUP	Technology	Spain	0.905
ENGIE	Utilities	France	0.886
BCO BILBAO VIZCAYA ARGENTARIA	Banks	Spain	0.885
TELEFONICA	Telecommunications	Spain	0.857
VIVENDI	Consumer	France	0.855
BMW	Automobiles	Germany	0.788
FRESENIUS	Healthcare	Germany	0.785
GRP SOCIETE GENERALE	Banks	France	0.547

Notes: This table reports the constituents of the EURO STOXX 50 index as of June 30th 2020. It should be noted that the index is subject to additions and deletions, which is not accounted for in this table. Instead, the table provides an illustrative overview of the types of companies included at the end of our sample period.

8.5 Hyperparameter Tuning

Table 9: Tuning of Hyperparameters

Panel A: Regularized Linear Models (RR, LA, EN)			
	RR	**LA**	**EN**
Penalty term (λ)	$[10^{-5}, 10^{5}]^*$	$[10^{-10}, 10^{1}]^*$	$[10^{-5}, 10^{2}]^*$
Mixing parameter (α)	–	–	$\{0.01, \ldots, 1.00\}^*$

Panel B: Tree-Based Models (BG, RF, GB)			
	BG	**RF**	**GB**
Minimum node size	5	5	–
Number of trees	500	500	$\{50, 100, \ldots, 500\}^*$
Feature split (m)	–	$p/3$	–
Tree depth	–	–	$\{1, 2\}^*$
Learning rate	–	–	$\{0.01, 0.1\}^*$

Panel C: Neural Networks (NN)	
Learning rate	0.001
Epochs	500
Patience	100
Drop-out	0.8
Initializer	Glorot normal
Optimizer	Adam (default)
Ensemble size	$\{1, 10\}$

Notes: This table reports the hyperparameter choices and tuning grids used across model classes. Parameters marked with an asterisk are selected via validation-set tuning. RR, LA, and EN denote ridge, lasso, and elastic net regression, respectively. BG, RF, and GB denote bagging, random forest, and gradient boosting models. p denotes the number of predictors in the corresponding feature set. The full mixing-parameter grid for EN is $\{0.01, 0.05, 0.10, 0.20, 0.30, 0.40, 0.50, 0.70, 0.90, 1.00\}$.

8.6 Description of Exogenous Variables

Table 10: Description of Predictor Variables in the $\mathcal{M}_{\text{HAR}}$ and $\mathcal{M}_{\text{ALL}}$ Feature Sets

Variable	Abbreviation	Description
A. HAR Model		
Daily Realized Variance	RVD	Previous-day realized variance
Weekly Realized Variance	RVW	Trailing 5-day average of realized variance
Monthly Realized Variance	RVM	Trailing 22-day average of realized variance
B. LevHAR Extensions		
Daily Negative Return	RD^-	Negative daily return component
Weekly Negative Return	RW^-	Trailing 5-day average of negative returns
Monthly Negative Return	RM^-	Trailing 22-day average of negative returns
C. SHAR Extensions		
Positive Semivariance	RVD^+	Previous-day variation from positive returns
Negative Semivariance	RVD^-	Previous-day variation from negative returns
D. HARQ Extension		
Realized Quarticity	RQ	Time-varying proxy for noise in RVD
E. Macroeconomic Variables		
VSTOXX Index	$VSTOXX$	Implied volatility (EURO STOXX 50)
Economic Policy Uncertainty	EPU	Policy uncertainty index
EUR/USD Exchange Rate	EUR/USD	Daily log return of the EUR/USD rate
Composite Indicator of Systematic Stress	$CISS$	ECB financial stress indicator
Brent Crude Oil Price	$BRENT$	Europe Brent crude oil spot price
Nikkei 225 Return	$NIKKEI_{ret}$	Daily return of Japanese equity index
Nikkei 225 Squared Return	$NIKKEI_{sq}$	Proxy for volatility of Japanese equity index
S&P 500 Return	$SP\&500_{ret}$	Daily lagged return of US equity index
S&P 500 Squared Return	$SP\&500_{sq}$	Proxy for volatility of US equity index
One-week Momentum	$M1W$	1-week log return of EURO STOXX 50
One-month Momentum	$M1M$	1-month log return of EURO STOXX 50
Term Spread	$TERM$	10-year Euro swap vs. 3-month Euribor spread
Credit Spread	$CREDIT$	Corporate bond yield spread

Notes: This table provides a description of the variables used in the $\mathcal{M}_{\text{HAR}}$ and $\mathcal{M}_{\text{ALL}}$ feature sets. Variables are grouped according to model extensions. Panel A corresponds to the standard HAR model. Panel B contains additional variables used in the LevHAR specification, such that the LevHAR model is constructed using Panels A and B. Panel C contains the semivariance components used in the SHAR model, which combines Panels A and C. Panel D includes realized quarticity, used in the HARQ model, which combines Panels A and D. The LogHAR specification only relies on the variables in Panel A, with a logarithmic transformation applied to each realized variance measure. Panel E contains the set of macroeconomic and financial variables included in the extended $\mathcal{M}_{\text{ALL}}$ feature set. These variables are incorporated in the HAR-X model, as well as in the regularized and machine learning methods, which are all estimated using the combined set of predictors from Panels A and E.

Table 11: Description of Predictor Variables in the $\mathcal{M}_{\mathrm{HAR}}$ and $\mathcal{M}_{\mathrm{ALL}}$ Feature Sets

Variable	Abbreviation	Transformation	Source
A. MHAR Feature Set			
Daily Realized Variance	RVD	-	Own calculations
Weekly Realized Variance	RVW	-	Own calculations
Monthly Realized Variance	RVM	-	Own calculations
B. LevHAR Extensions			
Daily Negative Return	RD^{-}	-	Own calculations
Weekly Negative Return	RW^{-}	-	Own calculations
Monthly Negative Return	RM^{-}	-	Own calculations
C. SHAR Extensions			
Positive Semivariance	RVD^{+}	-	Own calculations
Negative Semivariance	RVD^{-}	-	Own calculations
D. HARQ Extension			
Realized Quarticity	RQ	-	Own calculations
E. Macroeconomic Variables			
VSTOXX Index	$VSTOXX$	Levels	Eurex / Datastream
Economic Policy Uncertainty	EPU	Logarithm	Baker et al. (2016)
EUR/USD Exchange Rate	EUR/USD	Daily % Change	Datastream
Systemic Stress Index	$CISS$	Levels	ECB
Brent Oil Price	$BRENT$	Daily % Change	Datastream
Nikkei 225 Return	$NIKKEI_{ret}$	Daily % Change	Datastream
Nikkei 225 Squared Return	$NIKKEI_{sq}$	Squared Returns	Own calculations
S&P 500 Return (lag)	$SP\&500_{ret}$	Daily % Change	Datastream
S&P 500 Squared Return (lag)	$SP\&500_{sq}$	Squared Returns	Own calculations
One-week Momentum	$M1W$	Log Return	Own calculations
One-month Momentum	$M1M$	Log Return	Own calculations
Credit Spread	$CREDIT$	First-difference	Datastream
Term Spread	$TERM$	First-difference	Datastream

Notes: This table provides a description of the variables used in the $\mathcal{M}_{\mathrm{HAR}}$ and $\mathcal{M}_{\mathrm{ALL}}$ feature sets. Variables are grouped according to model extensions. The transformation column reports the construction of each variable, including levels, logarithmic transformations, returns, squared returns, and first-differences. The label *Own calculations* refers to variables constructed from raw high-frequency intraday data obtained from the Stockholm School of Economics, which has been processed and subsequently transformed into each respective measures used for volatility forecasting.

8.7 Summary of Out-of-Sample Forecast Performance

Table 12: Loss Function Summary for $\mathcal{M}_{\mathrm{HAR}}$ Feature Set

	Loss Functions			
	MSE (10^{-8})	**QLIKE**	**MAE** (10^{-4})	**RMSE** (10^{-4})
HAR	5.66	0.24	0.53	2.38
HAR-X	5.66	0.24	0.53	2.38
LogHAR	5.30	0.20	0.46	2.30
LevHAR	5.39	0.42	0.55	2.32
SHAR	5.84	0.24	0.53	2.42
HARQ	4.79	0.19	0.44	2.19
RR	5.82	0.40	0.73	2.41
LA	5.89	0.42	0.77	2.43
EN	5.88	0.42	0.76	2.43
BG	5.33	0.19	0.48	2.31
RF	5.19	0.19	0.47	2.28
GB	5.51	0.34	0.63	2.35
$\mathbf{NN}_1^1$	5.47	0.33	0.59	2.34
$\mathbf{NN}_1^{10}$	5.71	0.29	0.57	2.39
$\mathbf{NN}_2^1$	5.63	0.29	0.56	2.37
$\mathbf{NN}_2^{10}$	5.65	0.33	0.60	2.38
$\mathbf{NN}_3^1$	5.69	0.39	0.67	2.39
$\mathbf{NN}_3^{10}$	5.68	0.35	0.62	2.38
$\mathbf{NN}_4^1$	5.70	0.28	0.55	2.39
$\mathbf{NN}_4^{10}$	5.92	0.36	0.64	2.43

Notes: This table reports the out-of-sample forecast performance of all 20 models estimated using the $\mathcal{M}_{\mathrm{HAR}}$ feature set. Forecast accuracy is evaluated using Mean Squared Error (MSE), Quasi-Likelihood (QLIKE), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). These loss functions are detailed further in Section (4.6). All measures are based on one-day ahead forecasts of realized variance over the test period from 2016-04-21 to 2020-08-06 and rounded to two decimals, where lower values indicate better predictive performance for all four loss functions. The MSE and QLIKE loss functions serve as the primary sources of performance evaluation in Section (5), whereas MAE and RMSE provide supplementary information.

Table 13: Loss Function Summary for $\mathcal{M}_{\text{ALL}}$ Feature Set

	Loss Functions			
	MSE (10^{-8})	**QLIKE**	**MAE** (10^{-4})	**RMSE** (10^{-4})
HAR	5.66	0.24	0.53	2.38
HAR-X	5.26	$403.88^{(*)}$	0.54	2.29
LogHAR	5.30	0.20	0.46	2.30
LevHAR	5.39	0.42	0.55	2.32
SHAR	5.84	0.24	0.53	2.42
HARQ	4.79	0.19	0.44	2.19
RR	5.22	$10.76^{(*)}$	0.63	2.29
LA	5.14	0.26	0.54	2.27
EN	5.14	0.26	0.54	2.27
BG	5.07	0.18	0.46	2.25
RF	4.82	0.18	0.44	2.20
GB	5.22	0.21	0.46	2.28
$\mathbf{NN}_1^1$	5.35	0.26	0.53	2.31
$\mathbf{NN}_1^{10}$	4.98	0.32	0.58	2.23
$\mathbf{NN}_2^1$	4.99	0.32	0.58	2.23
$\mathbf{NN}_2^{10}$	5.07	0.28	0.54	2.25
$\mathbf{NN}_3^1$	5.23	0.27	0.55	2.29
$\mathbf{NN}_3^{10}$	5.34	0.32	0.60	2.31
$\mathbf{NN}_4^1$	5.44	0.37	0.65	2.33
$\mathbf{NN}_4^{10}$	5.36	0.40	0.69	2.31

Notes: This table reports the out-of-sample forecast performance of all 20 models estimated using the $\mathcal{M}_{\text{ALL}}$ feature set. Forecast accuracy is evaluated using Mean Squared Error (MSE), Quasi-Likelihood (QLIKE), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). These loss functions are detailed further in Section (4.6). All measures are based on one-day ahead forecasts of realized variance over the test period from 2016-04-21 to 2020-08-06 and rounded to two decimals, where lower values indicate better predictive performance for all four loss functions. The MSE and QLIKE loss functions serve as the primary sources of performance evaluation in Section (5), whereas MAE and RMSE provide supplementary information. Values denoted with superscript $(*)$ are extreme outliers with respect to their QLIKE value. Both HAR-X and RR are both extreme outliers in this regard. We investigate this issue and confirm that the behavior stems from the fact that these models generate many negative forecasts during the test period, causing the negative forecasts to subsequently be replaced via the insanity filter and converted to the corresponding minimum realized variance value in the training set. The problem is that when such values are so small, such that they approach zero, and appear frequently in the forecast window, it causes the QLIKE score to explode, reflecting the loss function's strong asymmetric penalization of underestimations of next-day realized variance.

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